

Requirement Tracability Matrix					
Feature	Requirement ID	Requirement Description	Test case ID	Test case description	Status
Functional Requirements					
Facility & Event Updates	FEU-01	The system shall display the current open/closed and availability status of facilities.	TC_FEU_002	Verify system shows current open/closed times correctly	Deferred
			TC_FEU_003	Verify system shows availability status correctly	Passed
	FEU-02	The system shall aggregate and display upcoming events with the event name, time and location.	TC_FEU_004	Verify system display upcoming events with name, time and location correctly	Passed
	FEU-03	The system shall display near real-time data.	TC_FEU_005	Verify data updates without any delay	Passed
	FEU-04	The system shall display last retrieved data when live data cannot fetch.	TC_FEU_006	Verify last saved data is shown when no internet	Passed
Transit & Navigations	TN-01	The system shall display campus bus routes.	TC_TN_002	Verify bus routes are are displayed correctly for each bus	Passed
	TN-02	The system shall display arrival and departure time.	TC_TN_003	Verify arrival and departure times accurately displayed for each bus	Passed
	TN-03	The system shall provide updated schedules.	TC_TN_004	Verify system shows latest transit shedule updates	Failed
	TN-04	The system shall display current status such as on time, delayed or unavailable.	TC_TN_005	Verify bus status shows 'On Time' when no delay	Passed
			TC_TN_006	Verify bus status shows 'Delayed' when delay exists	Passed
			TC_TN_007	Verify bus status shows 'Unavailable' when no busses available	Passed
Club Hub	CH-01	The system shall let students to discover and join active communities.	TC_CH_002	Verify student can join a club successfully	Failed
	CH-02	The system shall display active clubs, communities with details.	TC_CH_001	Verify all clubs are shown with correct details	Failed
	CH-03	The system shall feature student-led groups like coding club, cybersecurity club and etc.	TC_CH_003	Verify student-led clubs available	Passed
	CH-04	The system shall display club activities including upcoming meetings and events.	TC_CH_004	Verify system display club activities correctly	Deferred
Check-in & Rewards System	CRS-01	The system shall provide a check-in button or action when user is at valid campus locations.	TC_CRS_001	Verify user can check-in at valid campus locations using QR scan	Blocked
	CRS-02	The system shall validate user location before awarding points	TC_CRS_002	Verify system validates location before awarding points	Blocked
	CRS-03	Upon a successful validated check-in, the system shall award a defined number of engagement points (e.g. +10 points) to the user's profile.	TC_CRS_003	Verify points are received after successful check-in	Blocked
	CRS-04	The system shall prevent a student from checking in at the same location more than once within a defined cool down period.	TC_CRS_005	Verify duplicate check-in is prevented	In Progress
	CRS-05	Key locations for check-in include university library, cafeteria, club meetings, event venues and transport.	TC_CRS_004	Verify check-in allowed only at valid campus locations	In Progress
	CRS-06	The system shall display students current points balance.	TC_CRS_006	Verify current points balance displayed correctly	Failed
	CRS-07	The system shall maintain and display a log of the student's past check-in events.	TC_CRS_007	Verify check-in history is available and shown	Failed
	CRS-08	The system shall implement logic to allow points to be redeemed against defined rewards.	TC_CRS_008	Verify rewards redemption works correctly	Deferred
	LOG-01	Verify registered user can log in with valid Student ID and Password	TC_LOG_STU_001	Verify successful login with correct id and password	Passed
	LOG-02	Verify user cannot log in with invalid credentials	TC_LOG_STU_014	Login with incorrect id and password	Passed
	LOG-03	Verify user profile (including points balance) loads correctly after login	TC_LOG_STU_002	Verify correct points balanace displayed after login	Failed

	User Login	LOG-05	Verify session expires after inactivity	TC_LOG_STU_006	Verify session expires after inactivity	Deferred
		LOG-06	Verify login page uses secure connection (HTTPS)	TC_LOG_STU_007	Verify login uses HTTPS secure connection	Deferred
		LOG-07	Verify no sensitive data (password, tokens) appears in URL or logs	TC_LOG_STU_008	Verify no sensitive data in URL or logs after login	Not Run
		LOG-08	Verify account lock after multiple failed login attempts	TC_LOG_STU_020	Verify account lock after multiple failed attempts	Deferred
		LOG-09	Verify password is case sensitive	TC_LOG_STU_004	Verify password is case sensitive	Passed
		LOG-10	Verify password is masked during entry	TC_LOG_STU_003	Verify password is masked	Passed
		LOG-11	Verify SQL injection is prevented in login fields	TC_LOG_STU_016	SQL injection in Student ID field	Passed
	TC_LOG_STU_017			SQL injection in Password field	Passed	
	User Registration	REG-01	Create a new student user profile with valid student ID and email	TC_REG_001	Verify student can register with all valid information	Failed
		REG-02	Attempt to create a duplicate profile (same student ID)	TC_REG_011	Verify duplicate student id rejected	Blocked
		REG-03	Create profile with missing required fields	TC_REG_007	Verify required field validation during registration	Blocked
		REG-04	Create profile with invalid email format	TC_REG_008	Verify invalid email format rejected	Blocked
		REG-05	After registration, verify data is encrypted/stored securely	TC_REG_005	Verify password is stored encrypted	Failed
		REG-06	After registration, verify user starts with 0 engagement points	TC_REG_004	Verify new student starts with 0 points	Blocked
	Non-Functional Reuirements					
	User Profiles & Authentication	UPA-01	The system must be able to create and manage student user profiles.	TC_UPA-001	Verify profile creation after registration	Failed
		UPA-02	The system must implement measures to ensure data integrity (e.g points balances are accurate and cannot be easily manipulated).	TC_UPA-002	Verify all data are accurate	Failed
				TC_UPA-003	Verify data cannot be manipulated	Blocked
		UPA-03	The system shall display the student's profile including their details	TC_UPA-004	Verify student can view their profiles	Failed
	Data Management & Security	DMS-01	All student data, including profile information and reward points balances, must be stored securely.	TC_DMS_001	Verify password stored securely (Weak storage cryptography)	Passed
		DMS-02	Student IDs must be masked in all display contexts. The system must not expose sensitive student data in logs or UI and protect from unauthorized users.	TC_DMS_002	Verify student ID is masked in all displayed content (Personally identifiable information leakage)	Failed
				TC_DMS_003	Verify sensitive data is not exposed (Sensitive data leakage)	Failed
				TC_DMS_004	Verify system is protected from unauthorized users (Broken Access Control)	Blocked
		DMS-03	The points balance must only be modifiable through server-side validated check-in logic, preventing client-side manipulation.	TC_DMS_005	Verify points balance cannot be manipulated (Integrity violation)	Blocked
	Reliability & Availability	RA-01	The system must handle all possible failures gracefully, without crashing or presenting unhandled exceptions to the user.	TC_RA_001	Verify system handles database failure without crash	Passed
		RA-02	In the event of a failed data feed, the application must display the most recently cached data.	TC_RA_002	Verify recently cached data is shown	Passed

Role	Test_ID	Description	Test Category	Input	Type	Pre conditions	Steps	Post conditions	Expected results	Actual results	Cycle1 Status (Initial)	Cycle2 Status (Regression)	Cycle3 Status (Regression)	Cycle4 Status (Regression)	Bug ID	Notes
Student	TC_LOG_STU_001	Verify successful student login with correct id and password	Functional	ID: STU_014 Password: dslark123pwd	Positive	Student must be registered	1. Open login page 2. Enter valid user id 3. Enter valid password 4. Click login button	User session active	Student redirect to the dashboard	Student successfully redirect to the dashboard	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STU_002	Verify correct points balance displayed after login	Functional	ID: STU_014 Password: dslark123pwd	Positive	Student has existing points	1. Login successfully 2. Click on points balance 3. Verify points balance is displayed	User session active	Correct points balance is displayed	Show a valid functional value (1450 as points balance for all users)	Fail	Fail	Fail	Fail	BUG_LOG_01	-
	TC_LOG_STU_003	Verify password is masked	Security - Data	Password: dslark123pwd	Positive	Login page is open	1. Type valid password in password field 2. Verify password shown as dots or asterisk	Password remains masked	Password character appear as dots or asterisk	Password remains masked properly	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STU_004	Verify password is case sensitive	Functional	ID: STU_014 Password: dslark123pwd Duplicate: dSLARK123pwd	Positive	Student registered with unique password	1. Login with lowercase letters 2. Login with uppercase letters 3. Login with correct password 4. Click login button	No session created for incorrect attempts	Only matching case format succeeds	Only matching case format succeeds	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STU_005	Verify generic error appears when login failed	Security - Config	ID: wrong234 Password: Wrongpass123	Positive	Student may or may not exist	1. Login with wrong id or password 2. Click error message	User remains on the login page	Generic error message displays without specifying the error	Login failed but no generic error message appears	Fail	Pass	Pass	Pass	BUG_LOG_02	Cycle2: Issue fixed. Display generic error message for student not implemented. Try again?
	TC_LOG_STU_006	Verify session expires after inactivity	Security - Auth	ID: STU_014 Password: dslark123pwd	Positive	Student is logged in	1. Login successfully 2. Leave site for defined timeout 3. Try to access protected page	Session invalidated	User redirected to login and requires re-authentication	-	Not Implemented	Not Implemented	Deferred	Deferred	-	Cycle1: Session timeout for staff not implemented Postponed due to time limitation
	TC_LOG_STU_007	Verify login uses HTTPS secure connection	Security - Config	URL: http://localhost:8081/login-secure.html	Positive	Login page is accessible	1. Check browser URL bar 2. Inspect network traffic	Page loaded securely	Application should use HTTPS protocol for all authentication requests	-	Not Applicable	Not Applicable	Not Applicable	Not Applicable	-	Cycle1: Application is running on local environment using HTTP (localhost:8081) HTTP is not configured in current setup
	TC_LOG_STU_008	Verify no sensitive data in URL or logs after login	Security - Data	ID: STU_014 Password: dslark123pwd	Positive	Student logs in successfully	1. Check browser history after login 2. Check server logs	Session active	No password or token appears in URL or logs after login	No sensitive data appears in URL or logs after login	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STU_009	Verify successful student login redirect to student dashboard	Authorization	ID: STU_014 Password: dslark123pwd	Positive	Student logs in successfully	1. Verify student login redirects to student dashboard correctly	User session active	Student redirects to the student dashboard	Student redirects to the student dashboard correctly	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STU_010	Verify if field is required	Functional	Student ID: empty Password: dslark123pwd	Negative	Student is on the login page	1. Leave id field empty 2. Enter valid password 3. Click login button	No session created	Login denied. Error message appear	Login denied. Display "Please enter both your Student ID and Password" error message	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STU_011	Verify password field is required	Functional	Student ID: STU_014 Password: empty	Negative	Student is on the login page	1. Enter valid user id 2. Leave password field empty 3. Click login button	No session created	Login denied. Error message appear	Login denied. Display "Please enter both your Student ID and Password" error message	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STU_012	Login with correct id and incorrect password	Functional	Student ID: STU_014 Password: Wrongpass123	Negative	Student must be registered	1. Open login page 2. Enter correct user id 3. Enter invalid password 4. Click login button	Login failed	User login denied and shows an error message	User login failed. But no error message displayed	Fail	Pass	Pass	Pass	BUG_LOG_03	Cycle2: Issue fixed. Display "Invalid credentials. Try again?"
	TC_LOG_STU_013	Login with incorrect id and correct password	Functional	Student ID: wrong234 Password: Wrongpass123	Negative	Student must be registered	1. Open login page 2. Enter invalid user id 3. Enter valid password 4. Click login button	Login failed	User login denied and shows an error message	User login failed. But no error message displayed	Fail	Pass	Pass	Pass	BUG_LOG_03	Cycle2: Issue fixed. Display "Invalid credentials. Try again?"
	TC_LOG_STU_014	Login with incorrect id and password	Functional	ID: wrong234 Password: Wrongpass123	Negative	Student must be registered	1. Open login page 2. Enter invalid user id 3. Enter invalid password 4. Click login button	Login failed	User login denied and shows an error message	User login failed. But no error message displayed	Fail	Pass	Pass	Pass	BUG_LOG_03	Cycle2: Issue fixed. Display "Invalid credentials. Try again?"
	TC_LOG_STU_015	Login with empty fields	Functional	Student ID: empty Password: empty	Negative	Student must be registered	1. Open login page 2. Leave id empty 3. Leave password empty 4. Click login button	Login failed	User login denied and shows an error message	User login failed. Error message appears	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STU_016	SQL injection in ID field	Security - Input	Student ID: 'OR '1'='1' Password: dslark123pwd	Negative	Login page loaded	1. Open login page 2. Enter 'OR '1'='1' as id field 3. Enter correct password 4. Click login button	No session created	User login denied	User login failed	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STU_017	SQL Injection in Password field	Security - Input	Student ID: STU_014 Password: 'OR '1'='1'	Negative	Login page loaded	1. Open login page 2. Enter valid user id 3. Enter SQL injection payload 4. Click login button	No session created	User login denied	User login failed	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STU_018	XSS in ID field	Security - Input	ID: <script>alert('XSS')</script> Password: Wrongpass123	Negative	Login page is open	1. Enter XSS payload in user id field 2. Enter any password 3. Click login button	No alert popup	Script not executed	System did not crash. Script not executed. No visible error message displayed	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STU_019	XSS in password field	Security - Input	ID: STU_014 Password: 	Negative	Login page is open	1. Enter valid user id 2. Enter XSS payload in password field 3. Click login button	No alert popup	Script not executed	System did not crash. Script not executed. No visible error message displayed	Pass	Pass	Pass	Pass	-	-
Staff	TC_LOG_STU_020	Brute force protection	Security - Auth	ID: STU_014 Password: Wrongpass123 (5 times) Then correct password dslark123pwd	Negative	Student account exists	1. Attempt to login 5 times with wrong password 2. Attempt 6th time with correct password	Account locked for 5-15 minutes	Account locked after 5 fail attempts. 6th fails	6th attempt succeeds without account locking	Fail	Fail	Fail	Deferred	BUG_LOG_05	Postponed due to time limitation
	TC_LOG_STU_021	Very long password	Security - Input	ID: STU_014 Password: A * 2000	Negative	Login page is open	1. Enter valid user id 2. Enter 2000 character password 3. Click login button	No Server error or crash	System rejects without crashing. Invalid input correctly	User login failed without crashing. Missing validation	Fail	Fail	Fail	Deferred	BUG_LOG_04	Postponed due to time limitation
	TC_LOG_STU_022	Verify sensitive details (passwords) are not repeated and are masked during login attempt	Security - Auth	ID: STU_014 Password: dslark123pwd	Negative	Login page is open. Developer tools is open on network tab	1. Enter valid student details 2. Login with correct password 3. Check network tab for request and response 4. Check payload tab	Login attempt recorded and intercepted password visible in network payload tab	The password field value is masked before transmission or the request body is encrypted or base64 encoded over a secure HTTPS connection	The login credentials travel across the network in plain text (password: Student1234567). The application lacks HTTPS encryption or frontend masking	Fail	Fail	Fail	Deferred	BUG_LOG_05	Postponed due to time limitation
	TC_LOG_STU_023	Verify response does not return sensitive student data over an unencrypted connection	Security - Auth	ID: STU_014 Password: dslark123pwd	Negative	Login page is open. Developer tools is open on network tab	1. Enter valid student details 2. Click login button 3. Check network request in dev tools 4. Check payload tab 5. Open application logs 6. Check localStorage details	New encrypted JWT token, username and role appear in network payload tab or application logs	JWT token, username and role appear in network tab and transmitted over an unencrypted HTTP connection	JWT token, username, and role is exposed in network tab and transmitted over an unencrypted HTTP connection	Fail	Fail	Fail	Deferred	BUG_LOG_06	Cycle2: One token promise we not display any response data. But token expired. Data found in browser's localStorage in application Postponed due to time limitation
	TC_LOG_STU_024	Verify backend rejects manipulated student login payload	Security - Auth	ID: STU_014 Password: dslark123pwd	Negative	Login page is open. Developer tools is open on console tab	1. Copy successful login request as JSON from network tab 2. Paste it in console tab 3. Change username to different email 4. Keep the password 5. Press enter and observe the result	Current user session stays active on dashboard. They're blocked and backend return 403 Forbidden error	The server must validate inputs for specific id and reject manipulated request with 403 Forbidden or 401 Unauthorized	Backend successfully handles modified request and returns 403 Forbidden status	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STF_01	Verify successful staff login with correct id and password	Functional	ID: STF_001 Password: hshark123M	Positive	User must be registered	1. Open login page 2. Enter valid user id 3. Enter valid password 4. Click login button	Staff user session active	User redirect to the dashboard	User successfully redirect to the dashboard	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STF_02	Verify password is masked	Security - Data	Password: hshark123M	Positive	Login page is open	1. Type valid password in password field 2. Verify password shown as dots or asterisk	Password remains masked	Password character appear as dots or asterisk	Password remains masked properly	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STF_03	Verify password is case sensitive	Functional	ID: STF_001 Password: hshark123M Duplicate: HSHARK123M	Positive	User registered with unique password	1. Login with lowercase letters 2. Login with uppercase letters 3. Login with correct password 4. Click login button	No session created for incorrect attempts	Only matching case format succeeds	Only matching case format succeeds	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STF_04	Verify generic error appears when login failed	Security - Config	ID: wrong234 Password: Wrongpass123	Positive	User may or may not exist	1. Login with wrong id or password 2. Click error message	User remains on the login page	Generic error message displays without specifying the error	Login failed but no generic error message appears	Fail	Pass	Pass	Pass	BUG_LOG_02	Cycle2: Issue fixed. Display generic error message for student not implemented. Try again?
	TC_LOG_STF_05	Verify session expires after inactivity	Security - Auth	ID: STF_001 Password: hshark123M	Positive	User is logged in	1. Login successfully 2. Leave site for defined timeout 3. Try to access protected page	Session invalidated	User redirected to login and requires re-authentication	-	Not Implemented	Not Implemented	Deferred	Deferred	-	Cycle1: Session timeout for staff not implemented Postponed due to time limitation
	TC_LOG_STF_06	Verify login uses HTTPS secure connection	Security - Config	URL: http://localhost:8081/login-secure.html	Positive	Login page is accessible	1. Check browser URL bar 2. Inspect network traffic	Page loaded securely	Application should use HTTPS protocol for all authentication requests	-	Not Applicable	Not Applicable	Not Applicable	Not Applicable	-	Cycle1: Application is running on local environment using HTTP (localhost:8081) HTTP is not configured in current setup
	TC_LOG_STF_07	Verify no sensitive data in URL or logs after login	Security - Data	ID: STF_001 Password: hshark123M	Positive	User logs in successfully	1. Check browser history after login 2. Check server logs	Session active	No password or token appears in URL or logs after login	No sensitive data appears in URL or logs after login	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STF_08	Verify successful staff login redirect to admin dashboard	Authorization	ID: STF_001 Password: hshark123M	Positive	User logs in successfully	1. Verify staff login redirects to admin dashboard correctly	User session active	Staff redirects to the admin dashboard	Staff also redirects to the student dashboard	Fail	Pass	Pass	Pass	BUG_LOG_07	Cycle1: Staff redirects to correct dashboard
	TC_LOG_STF_09	Verify if field is required	Functional	ID: empty Password: hshark123M	Negative	User is on the login page	1. Leave id field empty 2. Enter valid password 3. Click login button	No session created	Login denied. Error message appear	Login denied. Display "Please enter both your Student ID and Password" error message	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STF_10	Verify password field is required	Functional	ID: STF_001 Password: empty	Negative	User is on the login page	1. Enter valid user id 2. Leave password field empty 3. Click login button	No session created	Login denied. Error message appear	Login denied. Display "Please enter both your Student ID and Password" error message	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STF_11	Login with correct id and incorrect password	Functional	ID: STF_001 Password: Wrongpass123	Negative	User must be registered	1. Open login page 2. Enter valid user id 3. Enter invalid password 4. Click login button	Login failed	User login denied and shows an error message	User login failed. But no error message displayed	Fail	Pass	Pass	Pass	BUG_LOG_03	Cycle2: Issue fixed. Display "Invalid credentials. Try again?"
	TC_LOG_STF_12	Login with incorrect id and correct password	Functional	ID: wrong234 Password: hshark123M	Negative	User must be registered	1. Open login page 2. Enter invalid user id 3. Enter valid password 4. Click login button	Login failed	User login denied and shows an error message	User login failed. But no error message displayed	Fail	Pass	Pass	Pass	BUG_LOG_03	Cycle2: Issue fixed. Display "Invalid credentials. Try again?"
	TC_LOG_STF_13	Login with incorrect id and password	Functional	ID: wrong234 Password: Wrongpass123	Negative	User must be registered	1. Open login page 2. Enter invalid user id 3. Enter invalid password 4. Click login button	Login failed	User login denied and shows an error message	User login failed. But no error message displayed	Fail	Pass	Pass	Pass	BUG_LOG_03	Cycle2: Issue fixed. Display "Invalid credentials. Try again?"
	TC_LOG_STF_14	Login with empty fields	Functional	ID: empty Password: empty	Negative	User must be registered	1. Open login page 2. Leave id empty 3. Leave password empty 4. Click login button	Login failed	User login denied and shows an error message	User login failed. Error message appears	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STF_15	SQL injection in ID field	Security - Input	ID: 'OR '1'='1' Password: hshark123M	Negative	Login page loaded	1. Open login page 2. Enter 'OR '1'='1' as id field 3. Enter correct password 4. Click login button	No session created	User login denied	User login failed	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STF_16	SQL Injection in Password field	Security - Input	ID: STF_001 Password: 'OR '1'='1'	Negative	Login page loaded	1. Open login page 2. Enter valid user id 3. Enter SQL injection payload 4. Click login button	No session created	User login denied	User login failed	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STF_17	XSS in ID field	Security - Input	ID: <script>alert('XSS')</script> Password: hshark123M	Negative	Login page is open	1. Enter XSS payload in user id field 2. Enter any password 3. Click login button	No alert popup	Script not executed	System did not crash. Script not executed. No visible error message displayed	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STF_18	XSS in password field	Security - Input	ID: STF_001 Password: 	Negative	Login page is open	1. Enter valid user id 2. Enter XSS payload in password field 3. Click login button	No alert popup	Script not executed	System did not crash. Script not executed. No visible error message displayed	Pass	Pass	Pass	Pass	-	-
	TC_LOG_STF_19	Brute force protection	Security - Auth	ID: STF_001 Password: Wrongpass123 (5 times) Then correct password hshark123M	Negative	User account exists	1. Attempt to login 5 times with wrong password 2. Attempt 6th time with correct password	Account locked for 5-15 minutes	Account locked after 5 fail attempts. 6th fails	6th attempt succeeds without account locking	Fail	Fail	Fail	Deferred	BUG_LOG_05	Postponed due to time limitation
	TC_LOG_STF_20	Very long password	Security - Input	ID: STF_001 Password: A * 2000	Negative	Login page is open	1. Enter valid user id 2. Enter 2000 character password 3. Click login button	No Server error or crash	System rejects without crashing. Invalid input correctly	User login failed without crashing. Missing validation	Fail	Fail	Fail	Deferred	BUG_LOG_04	Postponed due to time limitation

		TC_LOG_STP_21	Verify sensitive staff details (passwords) are not exposed and encrypted during high access network layer	Security - Auth	ID: STP_001 Password: hashem1234	Negative	Login page is open. Developer tools is open on network tab. Presence log enabled	1. Enter valid staff details 2. Click login button 3. Click high network request in dev tools 4. Check payload tab	Login attempt recorded and unencrypted password visible in network payload tab	The password field value should be masked or encrypted before transmission or the request must be strictly enforced over a secure HTTPS protocol	The login credentials travel across the network in plain text (password: "hashem1234"). The application lacks HTTPS encryption or frontend handling	Fail	Fail	Fail	Deferred	BUG_LOG_05	Postponed due to time limitation
		TC_LOG_STP_22	Verify response parameters do not expose sensitive staff data over an unencrypted connection	Security - Auth	ID: STP_001 Password: hashem1234	Negative	Login page is open. Developer tools is open on network tab. Presence log enabled and application localstorage	1. Enter valid staff details 2. Click login button 3. Click high network request in dev tools 4. Check preview tab 5. Open application dev 6. Check localstorage details	Raw encoded JWT token, username and role appears in network preview tab or application localstorage	JWT token, username and user role that is already encrypted or transmitted over secure HTTPS connection	JWT token, username, and role is exposed in network tab and transmitted over an unencrypted HTTP connection	Fail	Fail	Fail	Deferred	BUG_LOG_06	Check: Dev tools preview tab not displays any response data. No trace sensitive data found in browser's localstorage in application Postponed due to time limitation
		TC_LOG_STP_23	Verify backend rejects manipulated and login payloads	Security - Auth	ID: STP_001 Password: hashem1234	Negative	Login page is open. Developer tools is open on console tab	1. Copy successful login request as fetch command from network tab 2. Paste it in console tab 3. Change username to different username 4. Keep the password 5. Press enter and observe the result	Current user session stays active on dashboard. Request blocked and backend returns 403 Forbidden error	The server must validate details for specific id and reject unauthenticated request with 403 Unauthorized or 403 Forbidden error	Backend successfully handles modified request and returns 403 Forbidden status	Pass	Pass	Pass	Pass	-	-

Rule	TC_ID	Description	Test Category	Inputs	Type	Pre-conditions	Steps	Post-conditions	Expected results	Actual results	Cycle1 Status (Initial)	Cycle2 Status (Regression)	Cycle3 Status (Regression)	Cycle4 Status (Regression)	Bug ID	Notes
	TC_REQ_001	Verify student can register with all valid information	Functional	First name: John Last name: Silva ID: STU_018 Email: john@gmail.com Phone: 0773647468 Enrolled year: 2020 Department id: D001 Password: Password@123	Positive	Student must not be registered	1. Enter valid first name 2. Enter valid last name 3. Enter valid student id 4. Enter valid email address 5. Enter valid phone number 6. Enter strong password 7. Enter enrolled year 8. Enter department id 9. Click register button	Student details stored in database. Password stored securely.	Account created. Success message appear. Student redirected to the login page.	Registration successful. Success message appears	Pass	Fail	Fail	Fail	BUG_REQ_07	Cycle4: User strong attributes in both data
	TC_REQ_002	Verify system accepts valid student email format	Functional	First name: John Last name: Silva ID: STU_018 Email: john@gmail.com Phone: 0773647468 Enrolled year: 2020 Department id: D001 Password: Password@123	Positive	Registration page is open. Student must not be registered	1. Enter valid student details 2. Enter email in correct format 3. Click register button	System accepts email. Account created.	Registration allowed. Account created.	System accepted email. User registration successful.	Pass	Blocked	Blocked	Blocked	-	Cycle2: Testing could not be continue because main test case failed due to backend validation error
	TC_REQ_003	Verify strong password is accepted	Security - Auth	First name: John Last name: Silva ID: STU_018 Email: john@gmail.com Phone: 0773647468 Enrolled year: 2020 Department id: D001 Password: Password@123	Positive	Registration page is open	1. Enter valid student details 2. Register successfully 3. Log in as new student 4. Click register button	Student account created. Password stored securely.	Registration succeeds. Password accepted.	Password accepted and registration successful.	Pass	Blocked	Blocked	Blocked	-	Cycle2: Testing could not be continue because main test case failed due to backend validation error
	TC_REQ_004	Verify new student starts with 0 points	Functional	First name: John Last name: Silva ID: STU_018 Email: john@gmail.com Phone: 0773647468 Enrolled year: 2020 Department id: D001 Password: Password@123	Positive	Registration page is open	1. Enter new valid student details 2. Register successfully 3. Log in as new student 4. Observe points start with zero	Student has zero points	Points balance displayed as zero	Points balance shows 0.00/100.00 (value 1450) for all users after successfully registered	Fail	Blocked	Blocked	Blocked	BUG_REQ_01	Cycle2: Testing could not be continue because main test case failed due to backend validation error
	TC_REQ_005	Verify password is stored encrypted	Security - Data	First name: John Last name: Silva ID: STU_018 Email: john@gmail.com Phone: 0773647468 Enrolled year: 2020 Department id: D001 Password: Password@123	Positive	Registration page is open	1. Register student with valid strong password 2. Check database or API request and response 3. Verify password encrypted	Database contains encrypted password	Password is encrypted and not stored as plain text	Password is stored in encrypted form correctly.	Pass	Blocked	Blocked	Blocked	-	Cycle2: Testing could not be continue because main test case failed due to backend validation error
	TC_REQ_006	Verify password is masked	Security - Data	Password: Password@123	Positive	Registration page is open	1. Type valid password in password field 2. Verify password shown as dots or asterisk	Password remains masked	Password characters appear as dots or asterisk. Same for Confirm Password field	Password remains masked	Pass	Blocked	Blocked	Blocked	-	Cycle2: Testing could not be continue because main test case failed due to backend validation error
	TC_REQ_007	Verify required field validation during registration	Functional	First name: empty Last name: Smith ID: STU_017 Email: smith@yahoo.com Phone: empty Enrolled year: 2020 Department id: D005 Password: Password@123	Negative	Registration page is open	1. Leave 1 or 2 required fields as empty 2. Click register button	Student remain on registration page	Registration not allowed. Validation message should be displayed.	Registration failed.	Pass	Blocked	Blocked	Blocked	-	Cycle2: Testing could not be continue because main test case failed due to backend validation error
Student	TC_REQ_008	Verify invalid email format rejected	Functional	First name: Taylor Last name: Smith ID: STU_017 Email: taylor.graciano@gmail.com Phone: 0725463378 Enrolled year: 2023 Department id: D005 Password: Password@123	Negative	Registration page is open	1. Enter valid student details 2. Enter email with incorrect format 3. Click register button	Student not created	Registration fails. Error message appear: "Enter valid email address"	Registration succeeds with invalid email address. No validation message appears	Fail	Blocked	Blocked	Blocked	BUG_REQ_02	Cycle2: Testing could not be continue because main test case failed due to backend validation error
	TC_REQ_009	Verify weak password rejected	Security - Auth	First name: Tony Last name: Clark ID: STU_018 Email: tony.bry@gmail.com Phone: 0702493052 Enrolled year: 2024 Department id: D004 Password: 123	Negative	Registration page is open	1. Enter valid student details 2. Enter a weak password 3. Enter matching password in confirm field 4. Click register button	Student not created	Registration fails. Error message appear: "Password must contain at least 8 characters"	Registration succeeds with a weak password. No validation message appears	Fail	Blocked	Blocked	Blocked	BUG_REQ_03	Cycle2: Testing could not be continue because main test case failed due to backend validation error
	TC_REQ_010	Verify password mismatch handled	Functional	First name: Selena Last name: Gomez ID: STU_019 Email: selena@gmail.com Phone: 0768076009 Enrolled Year: 2020 Department id: D004 Password: Password@123 Confirm password: Password@12	Negative	Registration page is open	1. Enter valid student details 2. Enter "password1" in password field 3. Enter different "password2" in confirm password field 4. Click register button	Student not created	Registration fails. Error message appear: "Passwords do not match"	Not Implemented	Not Implemented	Blocked	Blocked	Blocked	-	Cycle1: Cycle2: Confirm password field not implemented
	TC_REQ_011	Verify duplicate id is rejected	Functional	First name: Rita Last name: Chen ID: STU_016 Email: rita@gmail.com Phone: 0768007769 Enrolled Year: 2020 Department id: D005 Password: Password@123	Negative	User already exist with same student id	1. Enter already exist student id in id field 2. Enter new student details in other fields 3. Click register button	Student not created	Registration fails. Error message appear: "Student ID already registered"	Registration failed. But no error message is displayed	Fail	Blocked	Blocked	Blocked	BUG_REQ_04	Cycle2: Testing could not be continue because main test case failed due to backend validation error
	TC_REQ_012	Verify duplicate email is rejected	Functional	First name: Rita Last name: Chen ID: STU_020 Email: john@gmail.com Phone: 0705500770 Enrolled Year: 2020 Department id: D005 Password: Password@123	Negative	User already exist with same email	1. Enter valid student details 2. Enter already exist email address 3. Click register button	Student not created	Registration fails. Error message appear: "Email already registered"	Registration succeeds with already exist case email	Fail	Blocked	Blocked	Blocked	BUG_REQ_05	Cycle2: Testing could not be continue because main test case failed due to backend validation error
	TC_REQ_013	SQL injection in ID field	Security - Input	First name: Ariana Last name: Garcia ID: "OR "1"="1" Email: ariana.garcia@gmail.com Phone: 0767076690 Enrolled Year: 2021 Department id: D001 Password: Password@123	Negative	Login page loaded	1. Open login page 2. Enter "OR "1"="1" in id field 3. Enter valid details in other fields 4. Click login button	No session created	Input is accepted as a normal string. No data leak or crash	System console and accepts as a normal string	Pass	Blocked	Blocked	Blocked	-	Cycle2: Testing could not be continue because main test case failed due to backend validation error
	TC_REQ_014	SQL injection in email field	Security - Input	First name: Jonas Last name: Blue ID: STU_021 Email: "OR "1"="1"@gmail.com Phone: 0702333040 Enrolled Year: 2020 Department id: D005 Password: Password@123	Negative	Login page loaded	1. Open login page 2. Enter "OR "1"="1" in email field 3. Enter valid details in other fields 4. Click login button	No session created	Input is accepted as a normal string. No data leak or crash	System console and accepts as a normal string	Pass	Blocked	Blocked	Blocked	-	Cycle2: Testing could not be continue because main test case failed due to backend validation error
	TC_REQ_015	Verify response parameters do not expose sensitive student data over an unencrypted connection	Security - Auth	First name: Lahiru Last name: Senarathna ID: STU_022 Email: lahiru.senarathna@gmail.com Phone: 0717558466 Enrolled Year: 2023 Department id: D001 Password: Password@123	Negative	Signup page is open. Developer tools is open on network with "Preserve log" enabled	1. Enter valid student details 2. Click register button 3. Click signup network request in dev tools 4. Check payload tab	Signup record expose student's details	Registered details must be securely encrypted or transmitted over secure HTTPS connection	All student's details are exposed in network tab, and transmitted over an unencrypted HTTP connection	Fail	Blocked	Blocked	Blocked	BUG_REQ_06	Cycle2: Testing could not be continue because main test case failed due to backend validation error

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TC_FEU_001	Verify system shows facility details correctly	Functional	Positive	Valid facility details exist in database	1. Navigate to the Facility updates section 2. View facility details	-	Facility details displays correctly	As for facility details system displays raw JSON values	Not Implemented	Fail	Pass	Pass	BUG_FEU_01	Cycle1: Facility & Event section not developed yet Cycle3: Issues Fixed
TC_FEU_002	Verify system shows current open/closed times correctly	Functional	Positive	Valid open/closed times exist in database	1. Navigate to the Facility updates section 2. Verify displayed open/closed times	-	System displays open/closed time correctly	-	Not Implemented	Not Implemented	Deferred	Deferred	-	Postponed due to time limitation
TC_FEU_003	Verify system shows availability status correctly	Functional	Positive	Availability status exist in database	1. Navigate to the Facility updates section 2. Verify displayed availability status	-	System displays status correctly	System displays "Quiet", "Empty" statuses	Not Implemented	Blocked	Pass	Pass	-	
TC_FEU_004	Verify system display events with name and location correctly	Functional	Positive	Gather correct names, times and location for every event	1. Open facility and event updates section 2. Verify every event displays correct event details	-	Event details displays correctly for every event	System failed to fetch available events with details	Not Implemented	Fail	Fail	Fail	BUG_FEU_02	Cycle1: Facility & Event section not developed yet Cycle2: Displays details correctly except event title
TC_FEU_005	Verify data updates without any delay	Performance	Positive	Internet connection available	1. Open facility and event updates section 2. Refresh page 3. Check if shown data changed	-	Newest data appears immediately after refresh without any delay	Cannot verify immediate update because page refresh causes redirect issue	Not Implemented	Fail	Fail	Fail	BUG_FEU_03	Cycle1: Facility & Event section not developed yet Cycle2: Implemented but found a new bug
TC_FEU_006	Verify last saved data is shown when no internet	Functional	Positive	Previous data exist in cache	1. Disable internet connection 2. Open facility and event updates section 3. View displayed information	Re-enable internet connection	Without displaying any error, previously cached data is shown	Previously fetched data displays	Not Implemented	Not Implemented	Pass	Pass	-	Cycle1: Facility & Event section not developed yet Cycle2: No caching mechanism implemented.
TC_FEU_007	Verify system displays 'Closed' outside operating hours	Functional	Positive	Gather operating hours for each facility	1. Open facility and event updates section 2. Check status during and after operating hours	-	Gym status shows 'Open' during and 'Closed' after operating hours	-	Not Implemented	Not Implemented	Deferred	Deferred	-	Postponed due to time limitation
TC_FEU_008	No duplicate events displayed	Functional	Positive	Data source send same event twice	1. Navigate to Events section 2. Check for duplicate events	-	Events appears only once in the list	Database prevents duplicate entries	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Facility & Event section not developed yet
TC_FEU_009	Data refreshes within expected time	Performance	Positive	Data refresh interval set to 30 seconds	1. Change displayed data in database 2. Refresh the page 3. Observe facilities details again	-	Facilities details update within the defined refresh interval	Cannot verify page refresh causes redirect issue	Not Implemented	Blocked	Fail	Fail	BUG_FEU_03	Cycle1: Facility & Event section not developed yet
TC_FEU_010	Past events not displayed in upcoming events section	Functional	Positive	Gather few past events	1. Open facility and event updates section 2. View upcoming events list 3. Verify no past events shown	-	No past events appear in upcoming events section	Past events appeared in upcoming events section	Not Implemented	Blocked	Fail	Fail	BUG_FEU_04	Cycle1: Facility & Event section not developed yet
TC_FEU_011	Verify system handles facility status with unexpected value	Security - Input	Negative	Data source sends 'Unknown' or 'Maintenance' for facility status	1. Open facility and event updates section 2. View facility status 3. Observe displayed value	-	System displays original value or 'Status unavailable' without crashing	System displays original value without crashing	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Facility & Event section not developed yet
TC_FEU_012	Verify system handles missing location for event	Security - Input	Negative	Data source sends other details without location for an event	1. Open facility and event updates section 2. View the event 3. Check if location available	-	System displays 'Location not available' or empty field without crashing	System displays original value without crashing	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Facility & Event section not developed yet

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	TC_TN_001	Verify Transit & Navigation section displays valid details	Functional	Positive	User logged in	1. Login to the application 2. Navigate to Transit & Navigation section 3. Verify displayed data	-	Bus details displays correctly according to the database	Failed to fetch database values	Not Implemented	Fail	Pass	Pass	BUG_TN_01	Cycle1: Data fetching not developed yet Cycle2: Failed to fetch Cycle3: Displays data correctly	
	TC_TN_002	Verify bus routes are displayed correctly for each bus	Functional	Positive	Gather current routes for each bus	1. Open Transit & Navigation section 2. Check routes of all listed buses with source	-	Bus routes are displayed correctly for each bus	Bus routes are displayed correctly for each bus	Not Implemented	Blocked	Pass	Pass		Cycle1: Data fetching not developed yet Cycle2: Cannot continue testing	
	TC_TN_003	Verify arrival and departure times accurately displayed for each bus	Functional	Positive	Gather arrival time and departure time for a chosen bus	1. Open Transit & Navigation section 2. Find details for chosen bus 3. Check arrival and departure time with source	-	Arrival and Departure times display accurately	Arrival and Departure times display accurately	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Data fetching not developed yet Cycle2: Cannot continue testing	
	TC_TN_004	Verify system shows latest transit schedule updates	Functional	Positive	Bus schedule updated 1 min ago	1. Open Transit & Navigation section 2. Refresh page 3. Check if data matches latest schedule	-	Most recent schedule data display	Cannot verify because refresh cause automatic redirection to profile page.	Not Implemented	Blocked	Fail	Fail	BUG_TN_02	Cycle1: Data fetching not developed yet Cycle2: Cannot continue testing	
	TC_TN_005	Verify bus status shows 'On Time' when no delay	Functional	Positive	Chosen bus has no delay	1. Open Transit & Navigation section 2. Find chosen bus from list 3. Check status	-	Bus status shows 'On Time' correctly	Bus status shows 'On Time' when no delay	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Data fetching not developed yet Cycle2: Cannot continue testing	
	TC_TN_006	Verify bus status shows 'Delayed' when delay exists	Functional	Positive	Chosen bus is delayed	1. Open Transit & Navigation section 2. Find chosen bus from list 3. Check status	-	Bus status shows 'Delayed' correctly	Bus status shows 'Delayed' when delay exist	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Data fetching not developed yet Cycle2: Cannot continue testing	
	TC_TN_007	Verify bus status shows 'Cancelled' when no buses available	Functional	Positive	Chosen bus is no available	1. Open Transit & Navigation section 2. Find chosen bus from list 3. Check status	-	Bus status shows 'Unavailable' correctly	Displays "Cancelled" when buses unavailable	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Data fetching not developed yet Cycle2: Cannot continue testing	
	TC_TN_008	Verify last cached data shown when offline	Functional	Positive	Previously bus data exists in cache	1. Disable internet connection 2. Open Transit & Navigation section 3. View displayed bus information	Re-enable internet connection	Last cached data is shown	-	Not Implemented	Not Implemented	Not Implemented	Not Implemented	-	Cycle1: Data fetching not developed yet Cycle2: Cannot continue testing Cycle3: No caching mechanism available	
	TC_TN_009	Verify system handles missing arrival time for a bus smoothly	Security - Input	Negative	Data source sends bus details without arrival time	1. Open Transit & Navigation section 2. View the bus details with missing arrival time 3. Check how arrival time field is displayed	-	Arrival time field shows as empty without crashing	Database added Not Null constraints to handle empty data properly	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Data fetching not developed yet Cycle2: Cannot continue testing	
	TC_TN_010	Verify system handles missing departure time for a bus smoothly	Security - Input	Negative	Data source sends bus details without departure time	1. Open Transit & Navigation section 2. View the bus details with missing departure time 3. Check how departure time field is displayed	-	Departure time field shows as empty without crashing	Database added Not Null constraints to handle empty data properly	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Data fetching not developed yet Cycle2: Cannot continue testing	
	TC_TN_011	Verify system handles missing bus route smoothly	Security - Input	Negative	Data source sends bus details without route	1. Open Transit & Navigation section 2. View the bus details without route 3. Check how route is displayed	-	System displays 'Route unavailable' without crashing	Database added Not Null constraints to handle empty data properly	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Data fetching not developed yet Cycle2: Cannot continue testing	
	TC_TN_012	Verify system handles invalid time format without crashing	Security - Input	Negative	Data source sends bus details with invalid time format	1. Open Transit & Navigation section 2. View the bus details with invalid time 3. Check how time is displayed	-	System displays 'Time not available' or empty without crashing	Database automatically adjust time according to the data type	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Data fetching not developed yet Cycle2: Cannot continue testing	
	TC_TN_013	Verify system handles empty data source without crashing	Security - Input	Negative	Data source sends empty details	1. Open Transit & Navigation section 2. Observe displayed content	-	Without crashing, system displays 'Bus not Available'	Database added Not Null constraints to handle empty data properly	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Data fetching not developed yet Cycle2: Cannot continue testing	
	TC_TN_014	Verify system handles duplicate bus entries	Functional	Negative	Data source sends same bus twice	1. Open Transit & Navigation section 2. Check for any duplicate buses	-	Duplicate may appear but app does not crash	Database prevents duplicate entries	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Data fetching not developed yet Cycle2: Cannot continue testing	

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	TC_CH_001	Verify all clubs are shown with correct details	Functional	Positive	Gather each available clubs with details	1. Navigate to Club Hub section 2. View all clubs 3. Check each club's details	-	All clubs are displayed with correct information	Failed to fetch and display club data	Not Implemented	Fail	Fail	Fail	BUG_CH_01	Cycle1: Club Hub section not developed yet
	TC_CH_002	Verify student can join a club successfully	Functional	Positive	Student is not a member of chosen club	1. Navigate to Club Hub section 2. Find chosen club 3. Click 'Join' button 4. Confirm join action	Student is added as a member of chosen club	Success feedback appear 'Joined [club name] successfully'. Club appears in 'My Clubs' section	Join works and data stores correctly in database but failed to display joined club's name, joined date. Also system displays other students' joined clubs too	Not Implemented	Blocked	Fail	Fail	BUG_CH_02	Cycle1: Club Hub section not developed yet Cycle2: Failed to fetch data
	TC_CH_003	Verify students-led clubs available	Functional	Positive	System has available clubs	1. Open Club Hub section 2. Check which clubs appear	-	Clubs section includes student-led clubs	Clubs section includes student-led clubs	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Club Hub section not developed yet
	TC_CH_004	Verify system display club activities correctly	Functional	Positive	Chosen club has an activity	1. Navigate to Club Hub section 2. Find chosen club 3. View activities list	-	Activity is visible with name, time and location correctly	-	Not Implemented	Not Implemented	Deferred	Deferred	-	Postponed due to time limitation
	TC_CH_005	Verify student can leave a club successfully	Functional	Positive	Student is logged in. Student is currently a member of chosen club	1. Navigate to Club Hub section 2. Go to 'My Clubs' or chosen club 3. Click 'Leave Club' button 4. Confirm leave action	Student is removed from chosen club	Success feedback appear 'Left [club name] successfully'. Club no longer appears in 'My Clubs' section	Uses wrong HTTP method and caused error	Not Implemented	Blocked	Fail	Fail	BUG_CH_03	Cycle1: Club Hub section not developed yet Cycle2: Failed to fetch data
	TC_CH_006	Verify student cannot join same club twice	Functional	Positive	Student is logged in. Student is already a member of chosen club	1. Navigate to Club Hub section 3. Check if 'Join' button still visible and clickable	-	'Join' button is disabled or shows 'Already Joined'	Duplicate entries prevented but displays success message	Not Implemented	Blocked	Fail	Fail	BUG_CH_04	Cycle1: Club Hub section not developed yet Cycle2: Failed to fetch data
	TC_CH_007	Verify student can view clubs they've joined	Functional	Positive	Student is logged in. Student is already joined to few clubs	1. Navigate to Club Hub section 2. Find 'My Clubs' or similar section 3. View list of joined clubs	-	Already joined clubs appear in 'My Clubs' section	Joined club appears with other students' joined clubs	Not Implemented	Blocked	Fail	Fail	BUG_CH_02	Cycle1: Club Hub section not developed yet Cycle2: Failed to fetch data
	TC_CH_008	Verify each club include member count	Functional	Positive	Chosen club has 30 members	1. Navigate to Club Hub section 2. Find chosen club 3. Check if member count visible	-	Chosen club shows member count as '30 Members'	-	Not Implemented	Not Implemented	Deferred	Deferred	-	Postponed due to time limitation
	TC_CH_009	Verify system handles when no clubs available	Functional	Positive	No clubs are available in the system	1. Navigate to Club Hub section 2. Observe displayed content	-	System displays 'No clubs available at this time' or similar	Failed to notify users about clubs availability	Not Implemented	Blocked	Fail	Fail	BUG_CH_05	Cycle1: Club Hub section not developed yet Cycle2: Failed to fetch data
	TC_CH_010	Verify system handles join attempt for non-existent club	Security - Input	Negative	Student is logged in, chosen club does not exist	1. Attempting to join club with invalid id or via URL manipulation or API 2. Observe system response	-	System displays 'Club not found'	-	Not Implemented	Blocked	Blocked	Blocked	-	Cycle1: Club Hub section not developed yet Cycle2: Failed to fetch data Cycle3: Cannot manipulate url with a static html url
	TC_CH_011	Verify system handles missing club description smoothly	Security - Input	Negative	Data source sends club details without description	1. Navigate to Club Hub section 2. View the club with missing description 3. Check description field	-	Description field shows 'No description available' or empty without crashing	-	Not Implemented	Not Implemented	Deferred	Deferred	-	Postponed due to time limitation
	TC_CH_012	Verify system handles missing club meeting time smoothly	Security - Input	Negative	Data source sends club details without including meeting time	1. Navigate to Club Hub section 2. View the club with missing time 3. Check meeting time field	-	Meeting time field shows 'Time not available' or empty without crashing	-	Not Implemented	Not Implemented	Deferred	Deferred	-	Postponed due to time limitation
	TC_CH_013	Verify system handles duplicate club smoothly	Functional	Negative	Data source sends a club twice	1. Navigate to Club Hub section 2. Find all available clubs 3. Check for duplicate clubs	-	Both may appear but app does not crash	Database prevents duplicate entries	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Club Hub section not developed yet Cycle2: Failed to fetch data
	TC_CH_014	Verify system handles join club when club is at full capacity	Security - Business Logic	Negative	Student is logged in. Chosen club has maximum capacity of 10 members and already has 10 members	1. Navigate to Club Hub section 2. Find chosen club 3. Click 'Join' button	Student is not added as member	Error appear 'Club is full'	-	Not Implemented	Not Implemented	Deferred	Deferred	-	Postponed due to time limitation
	TC_CH_015	Verify system handles club with missing location	Security - Input	Negative	Data source sends club details without including location	1. Navigate to Club Hub section 2. View the club with missing location 3. Check location field	-	Location field shows 'Location not available' or empty without crashing	-	Not Implemented	Not Implemented	Deferred	Deferred	-	Postponed due to time limitation

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TC_CRS_001	Verify user can check-in at valid campus locations using QR scan	Functional	Positive	Student is logged in Student is at chosen valid location. QR code is available	1. Open check-in section 2. Open QR scanner 3. Scan QR in chosen location 4. Confirm check-in	Check-in record is created	System displays "Checked in at [location name]" Points awarded	-	Not Implemented	Not Implemented	Blocked	Blocked		Cycle1: Check-in system not implemented Cycle3: Button not working
TC_CRS_002	Verify system validates location before awarding points	Security - Business logic	Positive	Student is logged in Student scans QR code	1. Scan QR code 2. System checks if location is valid 3. Observe points awarding		System validates the location and award points	-	Not Implemented	Not Implemented	Blocked	Blocked		Cycle1: Check-in system not implemented
TC_CRS_003	Verify points are received after successful check-in	Functional	Positive	Student is logged in Student has 10 points currently Check-in awards 5 points	1. Open check-in page 2. Perform check-in at a valid location 3. Check points balance after check-in	Points balance updated	Points balance increases to 15	-	Not Implemented	Not Implemented	Blocked	Blocked		Cycle1: Check-in system not implemented
TC_CRS_004	Verify check-in allowed only at valid campus locations	Security - Business logic	Positive	Student is logged in	1. Try to check-in at valid locations 2. Try to check-in at invalid location		Check-in successful at valid locations. Check-in rejected at invalid locations	-	Not Implemented	Not Implemented	Blocked	Blocked		Cycle1: Check-in system not implemented
TC_CRS_005	Verify duplicate check-in is prevented	Security - Business logic	Positive	Student is logged in Student already checked in at chosen valid location	1. Try to check-in at chosen location again in same day 2. Scan QR code again 3. Observe system response	No second check-in recorded. No duplicate points awarded	Error appears "You've already checked in here today"	-	Not Implemented	Not Implemented	Blocked	Blocked		Cycle1: Check-in system not implemented
TC_CRS_006	Verify current points balance displayed correctly	Functional	Positive	Student has 150 points	1. Log in to the application 2. Navigate to profile or rewards section 3. View points balance		Points balance shows '150' correctly	Failed to display student's points balance. Displays earned reward points	Not Implemented	Not Implemented	Fail	Fail	BUG_CRS_01	Cycle1: Check-in system not implemented
TC_CRS_007	Verify check-in history is available and shown	Functional	Positive	Student has checked in at few valid locations	1. Navigate to Check-in history 2. View list of past check-ins 3. Verify dates and locations		All check-ins appear with correct dates and locations	System displays all history check-in history	Not Implemented	Not Implemented	Fail	Fail	BUG_CRS_02	Cycle1: Check-in system not implemented
TC_CRS_008	Verify rewards redemption works correctly	Functional	Positive	Student has 50 points Free Coffee reward costs 30 points	1. Navigate to rewards section 2. Select 'Free Coffee' reward 3. Click redeem 4. Confirm redemption	Points deducted	Success message appear "Reward redeemed". Points balance reduced to 20		Not Implemented	Not Implemented	Deferred	Deferred		Cycle1: Check-in system not implemented Cycle3: Postponed due to time limitation
TC_CRS_009	Verify different locations give correct points	Functional	Positive	Library check-in gives 10 points. Cafeteria check-in gives 5 points	1. Check-in at Library 2. Check reward points 3. Check-in at Cafeteria 4. Check received points		Library gives 10 points and Cafeteria gives 5 points		Not Implemented	Not Implemented	Blocked	Blocked		Cycle1: Check-in system not implemented
TC_CRS_010	Verify points persist after logout and login	Security - Data	Positive	Student has 200 points	1. Note points balance 2. Logout from application 3. Log back in 4. Verify points balance		Points balance shows as 200		Not Implemented	Not Implemented	Blocked	Blocked		Cycle1: Check-in system not implemented
TC_CRS_011	Verify student cannot exceed daily check-in limit	Security - Business logic	Positive	Student is logged in. Daily check-in limit 3. Student has already checked in 3 times today	1. Attempt 4th check-in at valid location 2. Scan valid QR code 3. Observe system response	No 4th check-in recorded. Points not awarded	Error appears "Daily check-in limit reached. Come back tomorrow"		Not Implemented	Not Implemented	Blocked	Blocked		Cycle1: Check-in system not implemented
TC_CRS_012	Verify insufficient points prevents redemption	Functional	Positive	Student has 5 points Chosen reward costs 20 points	1. Navigate to reward section 2. Select chosen reward from reward list 3. Click redeem	No points deducted Reward not awarded	Error appears "Insufficient points balance"		Not Implemented	Not Implemented	Blocked	Blocked		Cycle1: Check-in system not implemented
TC_CRS_013	Verify check-in counter resets after midnight	Functional	Positive	Student checked 3 times on Monday. Time is now Tuesday 12:01 AM	1. Attempt check-in on Tuesday 2. Scan valid QR code 3. Observe response	Check-in successful	Check-in successful		Not Implemented	Not Implemented	Blocked	Blocked		Cycle1: Check-in system not implemented
TC_CRS_014	Verify check-in fails with invalid QR code	Security - Input	Negative	Student is logged in	1. Open QR scanner 2. Scan invalid QR code 3. Observe response	No check-in recorded	Error appears "Invalid QR code"		Not Implemented	Not Implemented	Blocked	Blocked		Cycle1: Check-in system not implemented
TC_CRS_015	Verify check-in fails at invalid location	Security - Business logic	Negative	Student is logged in Student scans QR code from off-campus location	1. Scan QR code at invalid location 2. Attempt check-in	No check-in recorded	Check-in rejects. Error appears "Not a valid check-in location"		Not Implemented	Not Implemented	Blocked	Blocked		Cycle1: Check-in system not implemented
TC_CRS_016	Verify redemption fails with insufficient points	Functional	Negative	Student has 15 points Chosen reward costs 30 points	1. Navigate to reward section 2. Select chosen reward from reward list 3. Click redeem	Points balance remains as 15	Error appears "Insufficient points balance"		Not Implemented	Not Implemented	Blocked	Blocked		Cycle1: Check-in system not implemented
TC_CRS_017	Verify redemption fails for unavailable reward	Functional	Negative	Student has 100 points Chosen reward is currently unavailable	1. Navigate to reward section 2. Select chosen reward 3. Click if redeem button is enabled	No points deducted	Redeem button is disabled or shows "This reward currently unavailable"		Not Implemented	Not Implemented	Blocked	Blocked		Cycle1: Check-in system not implemented
TC_CRS_018	Verify check-in fails when database connection is down	Security - Config	Negative	Student is logged in Database connection is down	1. Scan valid QR code 2. Attempt check-in 3. Observe response	Restore database connection	Error appears "System unavailable". No points awarded		Not Implemented	Not Implemented	Blocked	Blocked		Cycle1: Check-in system not implemented

Test ID	Description	Test Category	Type	Pre-conditions	Steps	Post-conditions	Expected results	Actual results	Cycle1 Status (Initial)	Cycle2 Status (Regression)	Cycle3 Status (Regression)	Cycle4 Status (Regression)	Bug ID	Notes
TC_UPA-001	Verify profile creation after registration	Functional	Positive	Student is registered and logged in.	1. Navigate to profile section 2. Check if profile is created 3. Verify details shown correctly	-	Profile created successfully with registered details	Profile created with details	Blocked	Fail	Fail	Fail	BUG_UPA_03	Cycle1: Database integration for the profile not yet implemented
TC_UPA-002	Verify all data are accurate	Functional	Positive	Student profile contains name, email, bio	1. Navigate to profile section 2. View all displayed details 3. Verify them	-	All details are accurate	Cannot verify because some fields are missing	Blocked	Fail	Fail	Fail	BUG_UPA_04	Cycle1: Database integration for the profile not yet implemented
TC_UPA-003	Verify data cannot be manipulated	Security - Data	Positive	Student is logged in. Student A is viewing their own profile. Student B has a profile	1. Attempt to manipulate URL to access user B's profile 2. Attempt to send API requests to change user B's data 3. Observe system response	No unauthorized changes made	System blocks access. Returns 'Access Denied'		Blocked	Blocked	Blocked	Blocked	-	Cycle1.2: Data manipulation cannot verify with static test url. Need a running environment
TC_UPA-004	Verify student can view their profiles	Functional	Positive	Student is logged in. Student has a complete profile	1. Navigate to profile section 2. Observe displayed information	-	Student sees their own profile with details	Student can view their profile but partially displays data	Not Implemented	Fail	Fail	Fail	BUG_UPA_03	Cycle1: Profile exists as a static page and does not fetch dynamic data
TC_UPA-005	Verify student can update profile details	Functional	Positive	Student is logged in	1. Navigate to profile section 2. Edit few details 3. Save changes	Details updated in database	Success message appear 'Profile saved successfully'	Success message appears but details not update in database	Fail	Fail	Fail	Fail	BUG_UPA_01	Cycle2: Update not works due to enrolled year null being sent and failed to load department is to choose Cycle2: Update not works due to enrolled year null being sent
TC_UPA-006	Verify data persists after logout and login	Functional	Positive	Student updates name field	1. Update name field 2. Logout 3. Log back in 4. View profile name	-	Profile name still shows updated name correctly		Blocked	Blocked	Blocked	Blocked	-	Cycle1: Database integration for the profile not yet implemented Cycle2: Update not works due to enrolled year null being sent
TC_UPA-007	Verify profile picture can be uploaded	Functional	Positive	Student is logged in	1. Navigate to profile section 2. Click upload picture 3. Select valid image 4. Save changes	Profile picture saved	Profile picture uploaded, saved and displays correctly	Change photo button lets pick an image but doesn't save it	Fail	Fail	Fail	Pass	BUG_UPA_02	Cycle2: Frontend failed to send backend required part
TC_UPA-008	Verify unauthorized user cannot edit another student's profile	Security - Auth	Negative	Student A is logged in. Student B has a profile	1. User A attempts to access user B's profile via URL manipulation 2. Observe system response	User B's data not exposed	Error message appears 'Access Denied'		Blocked	Blocked	Blocked	Blocked	-	Cycle1.2: Cannot test unauthorized access with static html url. Need a running environment
TC_UPA-009	Verify unauthorized user cannot edit another student's profile (Broken Object Level Authorization)	Security - Auth	Negative	Student A is logged in. Student A knows student B's student id	1. Student A attempts to send API requests to update student B's name 2. Observe system response	Student B's profile unchanged	API returns 403 error or 'Unauthorized'	Returns 403 Forbidden error	Pass	Pass	Pass	Pass	-	-
TC_UPA-010	Verify system rejects SQL injection in profile update fields. (Insecure Direct Object Reference)	Security - Input	Negative	Student is logged in	1. Navigate to profile section 2. Enter "OR 1=1" in name field 3. Attempt to save	No SQL injection executed	System prevents SQL injection		Blocked	Blocked	Blocked	Blocked	-	Cycle1: Database integration for the profile not yet implemented Cycle2: Profile update not work

Test_ID	Description	Test Category	Type	Pre-conditions	Steps	Post-conditions	Expected results	Actual results	Cycle1 Status (Init)	Cycle2 Status (Regression)	Cycle3 Status (Regression)	Cycle4 Status (Regression)	Bug ID	Notes
TC_RA_001	Verify system handles database failure without crash	Reliability	Positive	Application is running Student is logged in	1. Navigate to a section that reads from database (eg profile, check-in history) 2. Disconnect or stop database 3. Observe application behavior	Restores database connection	Application displays cached data or graceful errors without crashing	Application displays cash previous data without crashing	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Application currently uses static data for most features. Cannot verify database failure handling Cycle2: Cannot test, currently data not fetch from database
TC_RA_002	Verify recently cached data is shown	Reliability	Positive	Internet connection is enabled	1. Disable internet connection 2. Open all pages 3. Observe displayed data 4. Check if data reads matches previously loaded data	Re-enables internet connection	Most recently cached data is displayed	Previously fetched data is displayed	Not Implemented	Not Implemented	Pass	Pass	-	Cycle1: No caching mechanism implemented
TC_RA_003	Verify system recovers automatically when database connection established	Reliability	Positive	Database is down System currently shows cached data or error message	1. Observe current state 2. Restore database connection 3. Refresh the application 4. Observe data updates	Database connection restored	System automatically fetches new data from database	System automatically fetches new data from database	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Requires database failure and recovery logic Cycle2: Cannot test, currently data not fetch from database
TC_RA_004	Verify system handles partial database data	Reliability	Positive	Database returns error like event name and location	1. Navigate to events & updates section 2. Observe displayed events data 3. Check for any crashes or blank sections	-	Available data is displayed Missing fields handle gracefully without crashing	Database added Not Null constraints to handle empty data properly	Not Implemented	Blocked	Pass	Pass	-	Cycle1: Database integration for dynamic content is incomplete Cycle2: Cannot test, currently data not fetch from database
TC_RA_005	Verify system handles complete database errors	Reliability	Negative	Applications is running Database connection is disconnected	1. Simulate database outage 2. Attempt to log in 3. Attempt to check in 4. Attempt to view profile 5. Observe system behavior	Restores database connection	Systems does not crash or expose database errors	Systems does not crash previously fetched data shows	Blocked	Blocked	Pass	Pass	-	Cycle1: Database outage simulation cannot be tested. Check-in and profile page not implemented
TC_RA_006	Verify system handles 500 server error smoothly	Reliability	Negative	Server endpoint is configured to return HTTP 500 error	1. Navigate to section that fails periodically 2. Observe system response 3. Check if fetched data is shown	Remove 500 error simulation	System shows error message "Server error. Please try again later" without crashing	System shows error message "Server error. Please try again later" without crashing	Not Applicable	Not Applicable	Not Applicable	Not Applicable	-	Cycle1: 2: No server errors observed during testing
TC_RA_007	Verify system handles sudden network disconnection during operation	Reliability	Negative	Student is actively using the app	1. Perform check-in operation 2. During database request 3. Observe system response 4. Reconnect network 5. Verify no corrupted data	Reconnect network	System shows error message. No corrupted data saved	System shows error message. No corrupted data saved	Blocked	Blocked	Blocked	Blocked	-	Cycle1: 2: Check-in feature not implemented yet

Module	Test_ID	Description	Test Category	Inputs	Type	Pre-conditions	Steps	Post-conditions	Expected results	Actual results	Cycle1 Status (Initial)	Cycle2 Status (Regression)	Cycle3 Status (Regression)	Cycle4 Status (Regression)	Bug ID	Notes
Club Hub	TC_CH_STF_001	Successfully create a new club	Functional	Club Name: Music Club Status: true	Positive	Admin logged in	1. Enter club name 2. Click save	Club visible in database and table	"Club saved successfully" appears. New club appears in "Active Campus Societies Directory" table	Failed to save data	Not Implemented	Blocked	Fail	Fail	BUG_CH_STF_01	Cycle2: Failed to fetch data
	TC_CH_STF_002	Update existing club	Functional	Club Name: Music Club Status: false	Positive	"Music Club" already exist.	1. Change club status 2. Click save	Club updated in database and table	Club update correctly and appears updated details in table	-	Not Implemented	Blocked	Blocked	Blocked	-	Cycle2: Failed to fetch data
	TC_CH_STF_003	Delete existing club	Functional	-	Positive	"Music Club" already exist.	1. Click remove item in chosen club record	Club removed from table and database	Club removed successfully	Failed to remove due to foreign key constraints	Not Implemented	Blocked	Fail	Pass	BUG_CH_STF_03	Cycle2: Failed to fetch data
	TC_CH_STF_004	Create a club with duplicate name	Functional	Club Name: Music Club Status: false	Negative	-	1. Enter already exist club name 2. Enter club status 3. Click save	No duplicate club record created	Prevent duplicate entries. Displays "Club name already exist"	-	Not Implemented	Blocked	Blocked	Blocked	-	Cycle2: Failed to fetch data
	TC_CH_STF_005	Create club with empty name	Functional	Club Name: empty Status: false	Negative	-	1. Leave club name 2. Enter valid status 3. Click save	-	Displays "Club name required" or similar	-	Not Implemented	Blocked	Blocked	Blocked	-	Cycle2: Failed to fetch data
	TC_CH_STF_006	Create club with invalid status	Functional	Club Name: Dance Club Status: maybe	Negative	-	1. Save club with invalid status	-	Displays validation error	-	Not Implemented	Blocked	Blocked	Blocked	-	Cycle2: Failed to fetch data
Rewards Matrix	TC_RS_STF_001	Successfully create a new reward	Functional	Reward ID: DISC_5 Reward Name: 10% discount Points: 1000 Discount: 10	Positive	Admin logged in	1. Enter Reward ID 2. Enter Reward Name 3. Enter Points Required 4. Enter Discount Value % 5. Click save	Reward visible in database and table	"Reward saved successfully" appears. New reward appears in "Available Loyalty Rewards Items" table	"Reward listing published" displayed. New reward appeared after few seconds in "Available Loyalty Rewards Items" table	Not Implemented	Blocked	Pass	Pass	-	Cycle2: Failed to fetch check-in and rewards
	TC_RS_STF_002	Update existing reward	Functional	Reward ID: DISC_5 Reward Name: 10% discount Points: 1000 Discount: 10	Positive	DISC_5 reward exists	1. Click update on existing reward 2. Change Points from 1000 to 500 3. Click Save	-	Points update correctly and appears updated details in table	Update button not working	Not Implemented	Blocked	Fail	Fail	BUG_RS_STF_02	Cycle2: Failed to fetch check-in and rewards
	TC_RS_STF_003	Delete existing reward	Functional	-	Positive	DISC_5 reward exists	1. Click remove icon in chosen reward record	Reward removed from table and database	Reward removed successfully	Reward removed successfully from table and database	Not Implemented	Blocked	Pass	Pass	-	Cycle2: Failed to fetch check-in and rewards
	TC_RS_STF_004	Save a reward with duplicate ID	Functional	Reward ID: DISC_1	Negative	-	1. Enter already exist reward ID 2. Enter other fields properly 3. Click save	No duplicate reward record created	Prevent duplicate entries. Displays "Reward ID already exist"	System silently updated existing reward with new details	Not Implemented	Blocked	Fail	Fail	BUG_RS_STF_04	Cycle2: Failed to fetch check-in and rewards
	TC_RS_STF_005	Save reward with empty reward name	Functional	Reward ID: DISC_8 Reward Name: empty Points: 200 Discount: 10	Negative	-	1. Leave reward name 2. Click save	-	Displays "Reward name required" or similar	Creating failed correctly and display alert	Not Implemented	Blocked	Pass	Pass	-	Cycle2: Failed to fetch check-in and rewards
Facilities & Events	TC_FE_STF_001	Successfully create a new department	Functional	Dept ID: D005 Dept Name: Engineering	Positive	Admin logged in	1. Enter Department ID 2. Enter Department Title Name 3. Click save	Department visible in table	Displays "Department saved successfully" and new department appears in "Registered Campus Tracks Block" table	Displayed "Department successfully saved" and appeared in table after few seconds	Not Implemented	Blocked	Pass	Pass	-	-
	TC_FE_STF_002	Update existing department	Functional	Dept ID: D002 Dept Name: Aviation	Positive	Department "D002" exists	1. Select department record and click update 2. Enter new department name 3. Click save	Department updated in table	Displays "Department updated successfully" and updated data appears in "Registered Campus Tracks Block" table	Department name updated correctly in both table and database	Not Implemented	Blocked	Pass	Pass	-	-
	TC_FE_STF_003	Successfully create a new event	Functional	Event: Handmade Date: 2025-12-15 Time: 18:30 Venue: Car Park Committee: Robot Club	Positive	Department "D002" exists	1. Enter Event Name 2. Select Scheduled Date 3. Select Scheduled Time 4. Enter Venue 5. Enter Organizing Committee 6. Click save	Event visible in table	Success message appears. New event appears in "Staged Campus Events Calendar" table with unique ID	Failed to create a new event	Not Implemented	Blocked	Fail	Fail	BUG_FS_STF_01	wrong event title
	TC_FE_STF_004	Update existing event	Functional	Event: Handmade Date: 2025-12-15 Time: 17:30 Venue: Car Park Committee: Robot Club	Positive	Event "Handmade" exists	1. Enter new event time 2. Keep other data 3. Click save	Event updated in table	Displays "Event updated successfully" and updated data appears in "Staged Campus Events Calendar" table	Failed to update due to a database constraints failure	Not Implemented	Blocked	Fail	Fail	BUG_FS_STF_02	-
	TC_FE_STF_005	Delete existing event	Functional	-	Positive	-	1. Click remove icon in chosen event record	-	Delete successfully from the table and database	Successfully deleted from table and database	Not Implemented	Blocked	Pass	Pass	-	-
	TC_FE_STF_006	Create a new department with empty details	Functional	Dept ID: empty Dept Name: empty	Negative	-	1. Click save with empty fields	No new record created	Validation error like "Department ID and Name are required" appears	Displays "Please fill out both Department ID and Department Name"	Not Implemented	Blocked	Pass	Pass	-	-
	TC_FE_STF_007	Create a new department with duplicate ID	Functional	Dept ID: D005 Dept Name: Tourism	Negative	Department "D002" exists	1. Enter valid details in other fields 2. Enter duplicate department ID in ID 3. Click save	No new record created	Displays "Department ID already exist" or similar	Updated silently existing department without displaying error message	Not Implemented	Blocked	Fail	Fail	BUG_FS_STF_03	-
	TC_FE_STF_008	Create a new event with past date	Functional	Event: Isolation 3.0 Date: 2025-12-08 Time: 11:30 Venue: Auditorium Committee: Leo Club	Negative	-	1. Enter valid details in other fields 2. Fill other fields 3. Click save	-	Event not created with past date	-	Not Implemented	Blocked	Blocked	Blocked	-	-
	TC_FE_STF_009	Save event with empty required fields	Functional	Event: empty Date: 2025-11-05 Time: 11:30 Venue: Auditorium Committee: Leo Club	Negative	-	1. Leave event name as empty 2. Fill other fields 3. Click save	-	Failed to save event with missing required name	-	Not Implemented	Blocked	Blocked	Blocked	-	-
	TC_FE_STF_010	Save event with invalid date format	Functional	Event: Isolation 3.0 Date: 12/09/2028 Time: 11:30 Venue: Auditorium Committee: Leo Club	Negative	-	1. Enter valid details in other fields 2. Enter date with invalid format 3. Click save	-	Event not saved with invalid date format	-	Not Implemented	Blocked	Blocked	Blocked	-	-
	TC_FE_STF_011	Cancel deletion confirmation	Functional	Event: Handmade Date: 2025-12-15 Time: 17:30 Venue: Car Park Committee: Robot Club	Negative	-	1. Click remove icon in event record 2. Cancel confirmation dialog	-	Data remains in table	Data not deleted and remains	Not Implemented	Blocked	Pass	Pass	-	-
Location Control	TC_LC_STF_001	Successfully create a new location	Functional	Location ID: ALUX_01 Name: Main Campus Capacity: 1000	Positive	Admin logged in	1. Enter Location ID 2. Enter Location Name 3. Select Current Status 4. Click (Save Location)	Location visible with Live Operational Status	Success message appears and location appears in "Facilities Infrastructure Location Matrices" table	Success message appeared and location appeared in "Facilities Infrastructure Location Matrices" table	Not Implemented	Blocked	Pass	Pass	-	-
	TC_LC_STF_002	Update Location Status	Functional	-	Positive	Location exists	1. Click Actions (Edit) 2. Change Live Operational Status 3. Save	-	Status updated successfully	Edit action not working	Not Implemented	Blocked	Fail	Fail	BUG_LC_STF_01	-
	TC_LC_STF_003	Delete Location	Functional	-	Positive	Location exists with no dependencies	1. Click Delete	-	Location removed from table	Removed successfully	Not Implemented	Blocked	Pass	Pass	-	-
	TC_LC_STF_004	Save location with duplicate ID	Functional	Location ID: CYM_01 Name: Fitness Capacity: 100	Negative	-	1. Save location with duplicate location ID	-	Displays "Location ID already exist"	Updated silently existing location without displaying error message	Not Implemented	Blocked	Fail	Fail	BUG_LC_STF_02	-
	TC_LC_STF_005	Save capacity as a test	Functional	Capacity: infinite	Negative	-	1. Save location with Capacity as test	-	Validation error displays as "Capacity must be a number" or similar	Location not created. Failed to display error message	Not Implemented	Blocked	Fail	Fail	BUG_LC_STF_03	-
	TC_LC_STF_006	Save location with empty name	Functional	-	Negative	-	1. Save location with empty Location Name	-	Displays "Location Name is required"	Displays "Please fill out all fields"	Not Implemented	Blocked	Pass	Pass	-	-
Transit & Management	TC_LC_TM_001	Successfully create new bus	Functional	Bus ID: BUS_001 Bus Number: WP NB-5241 Status: ON TIME Departure: Campus Arrival: Main Campus Departure Time: 07:30 Arrival Time: 08:45	Positive	Admin logged in	1. Enter Bus ID 2. Enter Bus Number 3. Select Current Status 4. Select Departure Location 5. Select Arrival Location 6. Set Departure Time 7. Set Arrival Time 8. Click save	Bus visible with all details	Success message appears and bus appears in "Active Transit Inventory List" table	Failed to save data	Not Implemented	Blocked	Fail	Fail	BUG_TM_STF_01	-
	TC_LC_TM_002	Update bus status	Functional	Bus ID: BUS_001 Bus Number: WP NB-5241 Status: ON TIME Departure: Campus Arrival: Main Campus Departure Time: 07:30 Arrival Time: 08:45	Positive	Bus exists	1. Click Edit on existing bus 2. Change Status from ON_TIME to DELAYED 3. Save	-	Status updated in transit grid	-	Not Implemented	Blocked	Blocked	Blocked	-	-
	TC_LC_TM_003	Delete existing bus	Functional	-	Positive	Bus exists	1. Click Delete 2. Confirm	-	Bus removed from transit grid	Failed to remove	Not Implemented	Blocked	Fail	Fail	BUG_TM_STF_02	-
	TC_LC_TM_004	Save bus with duplicate bus ID	Functional	Bus ID: BUS_001	Negative	Bus "BUS_001" already exists	1. Save bus with duplicate Bus ID	-	Error "Bus ID already exists"	-	Not Implemented	Blocked	Blocked	Blocked	-	-
	TC_LC_TM_005	Save bus with empty bus number	Functional	-	Negative	-	1. Save bus with empty Bus Number	-	Error "Bus Number is required"	-	Not Implemented	Blocked	Blocked	Blocked	-	-
	TC_LC_TM_006	Save bus with invalid Status value	Functional	Status: FLYING (not in dropdown options)	Negative	-	1. Save bus with invalid Status value	-	Validation error or default to ON_TIME	-	Not Implemented	Blocked	Blocked	Blocked	-	-
	TC_LC_TM_007	Save bus with missing Departure Location	Functional	-	Negative	-	1. Save bus with missing Departure location	-	-	-	Not Implemented	Blocked	Blocked	Blocked	-	-
	TC_LC_TM_008	Save departure time with invalid format	Functional	Departure Time: 25:00	Negative	Bus exists	1. Edit bus 2. Set Departure Time to invalid format	-	Error "Invalid time format. Use HH:MM"	-	Not Implemented	Blocked	Blocked	Blocked	-	-

	Test ID	Testing Type	Dashboard	Test Scenario	Test Steps	Expected Result	Actual Result	Status
	TC_CHATBOT_01	Smoke	Both	Chatbot loads	1. Login 2. Open dashboard student/staff	Chatbot visible	Chatbot visible	Pass
	TC_CHATBOT_02	Smoke	Both	Send & Receive	1. Type "Hello" 2. Click send	Receives a reply	Displays Sorry, I'm having trouble connecting to the core server gateway right now.	Fail
	TC_CHATBOT_03	Smoke	Both	Factual Result	1. Ask how many kilometers from earth to moon	Reply contains 384,400 or 384,400km	Displays Sorry, I'm having trouble connecting to the core server gateway right now.	Fail
	TC_CHATBOT_04	Smoke	Both	API connectivity	1. Type any message	HTTP 200 response within 3 seconds	HTTP 200 response recieved within 3 seconds	Pass
	TC_CHATBOT_05	Functional	Both	Empty message	1. Click send with blank input	No API call OR "Please enter a message"	Displays Sorry, I'm having trouble connecting to the core server gateway right now.	Fail
	TC_CHATBOT_06	Functional	Both	Very long message	1. Type 1000+ characters	Response received (no timeout/crash)	Displays Sorry, I'm having trouble connecting to the core server gateway right now.	Fail
	TC_CHATBOT_07	Functional	Both	Special characters	1. Type "@*\$^\$^#&()%"	No crash, returns some response	Displays Sorry, I'm having trouble connecting to the core server gateway right now.	Fail
	TC_CHATBOT_08	Functional	Both	Non-English	1. Type "Bonjour" or "नमस्ते"	Responds appropriately (if supported)	Displays Sorry, I'm having trouble connecting to the core server gateway right now.	Fail
	TC_CHATBOT_09	Functional	Both	Numbers only	1. Type "647453"	Responds (not just echo the number)	Displays Sorry, I'm having trouble connecting to the core server gateway right now.	Fail
	TC_CHATBOT_10	Functional	Both	Follow-up	1. Ask "2+2?" then "3+3"	If stateful, understands about refers to addition	Displays Sorry, I'm having trouble connecting to the core server gateway right now.	Fail

	Feature	Bug ID	Bug Description	Test ID	Severity	Priority	Assign To	Status	Steps to Reproduce	Reported Date	Closed Date
	Login	BUG_LOG_01	Displays same static/hardcoded value (1450) for all users as points balance after login	TC_LOG_STU_002	Medium	High	Frontend Developer	Closed	1. Login to multiple students' account 2. Observe current points balance is same for all users'	15/05/2026	25/05/2026
Displays student earned reward's points. Failed to displays student earned points through check-in.			High		High	Frontend Developer	Open	1. Login using valid credentials 2. Observe displayed points is not earned points	22/05/2026		
		BUG_LOG_02	System does not displays error message for invalid login attempts	TC_LOG_STU_005	Medium	High	Frontend Developer	Closed	1. Login with invalid student Id and password 2. Observe error message not appears	15/05/2026	25/05/2026
				TC_LOG_STU_012	Medium	High	Frontend Developer	Closed	1. Login with valid student Id and invalid password 2. Observe error message not appears	15/05/2026	25/05/2026
				TC_LOG_STU_013	Medium	High	Frontend Developer	Closed	1. Login with invalid student Id and valid password 2. Observe error message not appears	15/05/2026	25/05/2026
				TC_LOG_STU_014	Medium	High	Frontend Developer	Closed	1. Login with invalid student Id and password 2. Observe error message not appears	15/05/2026	25/05/2026
				TC_LOG_STF_004	Medium	High	Frontend Developer	Closed	1. Login with invalid staff Id and password 2. Observe error message not appears	15/05/2026	25/05/2026
				TC_LOG_STF_011	Medium	High	Frontend Developer	Closed	1. Login with valid staff Id and invalid password 2. Observe error message not appears	15/05/2026	25/05/2026
				TC_LOG_STF_012	Medium	High	Frontend Developer	Closed	1. Login with invalid staff Id and valid password 2. Observe error message not appears	15/05/2026	25/05/2026
				TC_LOG_STF_013	Medium	High	Frontend Developer	Closed	1. Login with invalid staff Id and password 2. Observe error message not appears	15/05/2026	25/05/2026
		BUG_LOG_03	Account is not locked after multiple failed attempts	TC_LOG_STU_020	High	High	Backend Developer	Open	1. Try login as student with invalid password 5 times 2. Try login with valid password for 6th time 2. Observe account is not locked and login succeeds	15/05/2026	
				TC_LOG_STF_019	High	High		Open	1. Try login as staff with invalid password 5 times 2. Try login with valid password for 6th time 2. Observe account is not locked and login succeeds	15/05/2026	
		BUG_LOG_04	No input validation for character length in password	TC_LOG_STU_021	High	High	Frontend Developer & Backend Developer	Open	1. Try login with valid student Id and A"2000 as password 2. Observe no validation for input	15/05/2026	
				TC_LOG_STF_020	High	High		Open	1. Try login with valid staff Id and A"2000 as password 2. Observe no validation for input	15/05/2026	
		BUG_LOG_05	Password exposed in network payload tab	TC_LOG_STU_022	Critical	High	Frontend Developer	Open	1. Login with valid student details 2. Click login network request in dev tools 3. Observe password exposed in payload tab	15/05/2026	
				TC_LOG_STF_021	Critical	High		Open	1. Login with valid staff details 2. Click login network request in dev tools 3. Observe password exposed in payload tab	15/05/2026	
		BUG_LOG_06	JWT token, username, role exposed in browser's dev tools	TC_LOG_STU_023	Critical	High	Frontend Developer	Open	1. Login with valid student details 2. Click login network request in dev tools 3. Observe token, username and role exposed in preview tab or localstorage	15/05/2026	
				TC_LOG_STF_022	Critical	High		Open	1. Login with valid staff details 2. Click login network request in dev tools 3. Observe token, username and role exposed in preview tab or localstorage	15/05/2026	
		BUG_LOG_07	Staff redirects to same student dashboard. No separate admin dashboard exist	TC_LOG_STF_008	Critical	High	Frontend Developer & Backend Developer	Closed	1. Login with valid staff Id and password 2. Observe successful login redirects to student dashboard	15/05/2026	25/05/2026
		Registration	BUG_REG_01	New student's points balance not starts with zero	TC_REG_002	Critical	High	Backend Developer & Database Manager	Fixed	1. Register new student with valid details 2. Observe points balance not starts with zero	15/05/2026
	BUG_REG_02		No validation for invalid email format	TC_REG_008	High	High	Frontend Developer & Backend Developer	Open	1. Register new student with invalid email format 2. Observe registration succeeds with no error and validation	15/05/2026	
	BUG_REG_03		No validation for weak passwords	TC_REG_009	High	High	Backend Developer & Frontend Developer	Open	1. Register new student with weak password 2. Observe registration succeeds with no error and validation	15/05/2026	
	BUG_REG_04		No error message for registration with duplicat student Id	TC_REG_011	Medium	High	Backend Developer	Verified	1. Try to register new student with already exist Id 2. Observe registration failed but no error displays	15/05/2026	
	BUG_REG_05		No error message for registration with duplicat student email	TC_REG_012	Medium	High	Backend Developer	Verified	1. Try to register new student with already exist email 2. Observe registration failed but no error displays	15/05/2026	
	BUG_REG_06		All student's details exposed in network payload tab	TC_REG_016	Critical	High	Frontend Developer	Open	1. Signup with valid student details 2. Click signup network request in dev tools 3. Observe student details including id, password, email exposed in preview tab	16/05/2026	
	BUG_REG_07	Student registration fails due to null ID being sent from frontend	TC_REG_001	Critical	High	Frontend Developer	Open	1. Try to signup with valid student details 2. Click register button 3. Observe registration failed and console output in backend	22/05/2026		
	User Profiles & Authentication	BUG_UPA_01	Failed to store updated profile details in database	TC_UPA-005	Medium	Medium	Frontend Developer	Open	1. Update existing profile details 2. Click save changes 3. Observe feedback appears but details not stored	16/05/2026	
		BUG_UPA_02	Chosen image not saves as profile picture	TC_UPA-007	Low	Low	Frontend Developer	Closed	1. Click change photo 2. Choose preferred image 3. Observe image not saved	16/05/2026	25/05/2026
		BUG_UPA_03	Profile displays only few fields. Failed to display all registered data except ID	TC_UPA-001	Medium	Medium	Frontend Developer	Open	1. Navigate to profile section 2. Observe displayed data	22/05/2026	
		BUG_UPA_04	Profile details partially displayed. Cannot verify accuracy of missing fields	TC_UPA-002	Medium	Medium	Frontend Developer	Open	1. Navigate to profile section 2. Observe data in other fields missing	22/05/2026	
	Data Management & Security	BUG_DMS_01	Sensitive data exposed across browser's network payload and preview tabs	TC_DMS_003	Critical	High	Frontend Developer	Open	1. Navigate to network tab 2. Click login or signup request 2. Check payload and preview tabs 3. Observe sensitive data is displayed	17/05/2026	
		BUG_DMS_02	Frontend does not sanitise/escape malicious inputs before sending to server	TC_DMS_006	Low	Low	Frontend Developer	Open	1. Open login page with developer tools 2. In network tab, check Preserve log 3. Enter valid Id 4. Enter as password 5. Click login button 6. Click login request in network tab 7. Check payload tab 8. Observe input not sanitized and not rejected before sending	17/05/2026	
		BUG_DMS_03	No error messages displays to notify users	TC_DMS_010	High	High	Frontend Developer	Closed	1. Perform invalid login 2. Observe no error message appears	17/05/2026	25/05/2026

		BUG_DMS_04	Student Id displays in profile section but not properly masked	TC_DMS_002	Critical	High	Frontend Developer	Open	1. Open profile section 2. Observe student ID displayed without masking	25/05/2026	
	Facility & Event Updates	BUG_FEU_01	Facility details display as raw JSON and undefined values instead of user-friendly details	TC_FEU_001	High	High	Frontend Developer	Closed	1. Navigate to facility status section 2. Observe data displayed as raw JSON format	22/05/2026	25/05/2026
		BUG_FEU_02	Events section failed to fetch and display available events with details	TC_FEU_004	High	Medium	Frontend Developer	Open	1. Navigate to events section 2. Observe data failed to fetch and display correctly	22/05/2026	
		BUG_FEU_03	Facility statuses section refresh causes redirection to profile dashboard	TC_FEU_005	High	High	Frontend Developer	Open	1. Navigate to facility status section 2. Observe page refresh automatically redirect to profile dashboard	22/05/2026	
		BUG_FEU_04	Past events displays with upcoming events. Failed to display separately	TC_FEU_010	Low	Low	Frontend Developer	Open	1. Navigate to facility status section 2. Check each event's date 3. Observe past events and upcoming events display without seperating	23/05/2026	
	Transit & Navigation	BUG_TN_01	System failed to fetch and display transit bus details from the database	TC_TN_001	Critical	High	Frontend Developer	Closed	1. Navigate to transit and navigation section 2. Observe data failed to fetch and display correctly	22/05/2026	25/05/2026
		BUG_TN_02	Transit & Navigation section refresh causes redirection to profile dashboard	TC_TN_004	High	High	Frontend Developer	Open	1. Navigate to transit and navigation section 2. Observe page refresh automatically redirect to profile dashboard	23/05/2026	
	Club Hub	BUG_CH_01	System failed to fetch and display club details from the database	TC_CH_001	Critical	High	Frontend Developer	Closed	1. Navigate to clubs section 2. Observe data failed to fetch and display correctly	22/05/2026	25/05/2026
			Failed to display each club's status		High	High	Frontend Developer	Open	1. Navigate to clubs section 2. Observe active status	23/05/2026	
		BUG_CH_02	Failed to display student joined club's name and joined date	TC_CH_002	High	High	Frontend Developer	Open	1. Navigate to clubs section 2. Observe failed to display joined club's name, active status and joined date under My Enrolled Clubs section	23/05/2026	
					System displays other students's joined clubs as well	Critical	High	Frontend Developer	Open	1. Navigate to clubs section 2. Check displayed data with database under My Enrolled Clubs section	23/05/2026
		BUG_CH_STF_01	Failed to save new club data	TC_CH_STF_001	High	High	Frontend Developer	Open	1. Navigate to Club Management 2. Enter new club details 3. Click Save 4. Observe failure to save data	25/05/2026	
	Staff Dashboard	BUG_CH_STF_02	Student cannot leave a joined club due to wrong usage of HTTP method	TC_CH_STF_003	Medium	High	Frontend Developer	Open	1. Open joined clubs list 2. Attempt to leave a club 3. Request fails due to incorrect HTTP method usage	25/05/2026	
		BUG_CRS_STF_01	Update button in reward matrix not working	TC_RS_STF_002	High	High	Frontend Developer	Open	1. Open reward matrix 2. Click Update button	25/05/2026	
		BUG_CRS_STF_02	System silently updated existing reward with new details	TC_RS_STF_004	High	High	Frontend Developer	Open	1. Update reward details 2. Save changes	25/05/2026	
		BUG_FS_STF_01	Failed to create a new event	TC_FE_STF_003	High	High	Frontend Developer	Open	1. Navigate to Event creation 2. Fill event details 3. Click Create	25/05/2026	
		BUG_FS_STF_02	Failed to update due to a database constraints failure	TC_FE_STF_004	High	High	Frontend Developer	Open	1. Update an event 2. Trigger database constraint condition (e.g., duplicate/invalid data)	25/05/2026	
		BUG_FS_STF_03	Updated silently existing department without displaying error message	TC_FE_STF_007	Medium	High	Frontend Developer	Open	1. Edit department details in event module 2. Save changes	25/05/2026	
		BUG_LC_STF_01	Edit action in location control not working	TC_LC_STF_002	High	High	Frontend Developer	Open	1. Navigate to Location Control 2. Try editing a location 3. Save changes	25/05/2026	
		BUG_LC_STF_02	Updated silently existing location without displaying error message	TC_LC_STF_004	Medium	High	Frontend Developer	Open	1. Update location data 2. Save changes	25/05/2026	
		BUG_LC_STF_03	Location not created. Failed to display error message	TC_LC_STF_005	High	High	Frontend Developer	Open	1. Create a new location with invalid/duplicate data 2. Click save	25/05/2026	
		BUG_TM_STF_01	Failed to save transit data	TC_LC_TM_001	High	High	Frontend Developer	Open	1. Go to Transit Management 2. Enter bus/transit details 3. Click Save	25/05/2026	
		BUG_TM_STF_02	Failed to remove bus details	TC_LC_TM_003	High	High	Frontend Developer	Open	1. Open bus details list 2. Try to remove a bus entry 3. Deletion fails	25/05/2026	

Feature	Table Name	Test ID	HTTP Method	Endpoint	Test Description	Request Body	Expected Status	Cycle1 (Initial testing)	Cycle2 (Regression)	Cycle3 (Regression)	Bug ID	Notes
Database & Backend Connection	departments	TC_DBC_DEP_01	GET	/five-campus-hub/api/v1/departments	Get all departments	N/A	200 OK	Pass	Pass	Pass	-	-
		TC_DBC_DEP_02	GET	/five-campus-hub/api/v1/departments/D001	Get single department	N/A	200 OK	Pass	Pass	Pass	-	-
		TC_DBC_DEP_03	POST	/five-campus-hub/api/v1/departments	Create new department	{"department_id":"D001","department_Name":"Management"}	201 Created	Pass	Pass	Pass	BUG_DBC_DEP_01	Cycle2: Issue fixed
		TC_DBC_DEP_05	PUT	/five-campus-hub/api/v1/departments	Update exist department	{"department_id":"D001","department_Name":"Finance"}	204 No Content	Pass	Pass	Pass	-	-
		TC_DBC_DEP_06	DELETE	/five-campus-hub/api/v1/departments	Delete exist department	{"department_id":"D002","department_Name":"Engineering"}	204 No Content	Pass	Pass	Pass	-	Cycle 2: Added id to the endpoint
	students	TC_DBC_STD_01	GET	/five-campus-hub/api/v1/students	Get all students	N/A	200 OK	Fail	Pass	Pass	BUG_DBC_STD_01	Cycle 2: Fixed 'enrolled_year' data type mismatch
		TC_DBC_STD_02	GET	/five-campus-hub/api/v1/students/1	Get single student	N/A	200 OK	Fail	Pass	Pass	BUG_DBC_STD_02	Cycle2: Issue fixed
		TC_DBC_STD_03	POST	/five-campus-hub/api/v1/students	Create new student	{"student_id":"","52406234","student_pwd":"","2345671","first_name":"","Nimasha","last_name":"","Kaihan","phone":"","0715472527","email":"","nimashak@gmail.com","enrolled_year":"","2023","department_id":"","D002"}	201 Created	Fail	Pass	Pass	BUG_DBC_STD_03	Cycle2: Issue fixed
		TC_DBC_STD_04	POST	/five-campus-hub/api/v1/students/save-students	Create multiple students	[{"student_id":"","STU_013","student_pwd":"","student123#","first_name":"","Heyoon","last_name":"","Jeong","phone":"","0715472749","email":"","heyong@gmail.com","enrolled_year":"","2023","department_id":"","D001"}, {"student_id":"","STU_014","student_pwd":"","student123#","first_name":"","Sina","last_name":"","Deinet","phone":"","0715472500","email":"","deinet@gmail.com","enrolled_year":"","2024","department_id":"","D004"}]	201 Created	Blocked	Pass	Pass	-	New in Cycle 2 (Not available in Cycle 1)
		TC_DBC_STD_05	PUT	/five-campus-hub/api/v1/students/1	Update exist student	{"student_id":"","STU_013","student_pwd":"","sanduni123#","first_name":"","Sanduni","last_name":"","Hansika","phone":"","0714224503","email":"","sanduni@gmail.com","department_id":"","D004"}	204 No Content	Fail	Pass	Pass	BUG_DBC_STD_04	Cycle 2: Issue fixed but password turned into a plain text after update student's details
		TC_DBC_STD_06	DELETE	/five-campus-hub/api/v1/students/STU_014	Delete exist student	N/A	204 No Content	Fail	Pass	Pass	BUG_DBC_STD_05	Cycle2: Issue fixed
	rewards	TC_DBC_RWD_01	GET	/five-campus-hub/api/v1/rewards	Get all rewards	N/A	200 OK	Pass	Pass	Pass	-	-
		TC_DBC_RWD_02	GET	/five-campus-hub/api/v1/rewards/DISC_15	Get single reward	N/A	200 OK	Pass	Pass	Pass	-	-
		TC_DBC_RWD_03	POST	/five-campus-hub/api/v1/rewards	Create new reward	{"reward_id":"","DISC","reward_name":"","25% discount","reward_points":"","1080","discount_percentage":"","25"}	201 Created	Pass	Pass	Pass	BUG_DBC_RWD_01	Cycle2: Issue fixed
		TC_DBC_RWD_04	POST	/five-campus-hub/api/v1/rewards/add-points(points)	Add points to student	N/A	201 Created	Blocked	Pass	Pass	-	New in Cycle 2 (Not available in Cycle 1)
		TC_DBC_RWD_05	DELETE	/five-campus-hub/api/v1/rewards/DISC_5	Delete reward	N/A	204 No Content	Blocked	Pass	Pass	-	New in Cycle 2 (Not available in Cycle 1)
	locations	TC_DBC_LOC_01	GET	/five-campus-hub/api/v1/locations	Get all locations	N/A	200 OK	Pass	Pass	Pass	-	-
		TC_DBC_LOC_02	GET	/five-campus-hub/api/v1/locations/CAR_PARK	Get single location	N/A	200 OK	Pass	Pass	Pass	-	-
		TC_DBC_LOC_03	GET	/five-campus-hub/api/v1/locations/CAR_PARK/status	Get single location's status	N/A	200 OK	Blocked	Pass	Pass	-	New in Cycle 2 (Not available in Cycle 1)
		TC_DBC_LOC_04	GET	/five-campus-hub/api/v1/locations/all-status	Get location status	N/A	202 OK	Blocked	Pass	Pass	-	New in Cycle 2 (Not available in Cycle 1)
		TC_DBC_LOC_05	POST	/five-campus-hub/api/v1/locations	Create new location	{"location_id":"","GYM_01","location_name":"","Gym","capacity":"","500"}	201 Created	Pass	Pass	Pass	BUG_DBC_LOC_01	Cycle2: Issue fixed
		TC_DBC_LOC_06	PUT	/five-campus-hub/api/v1/locations	Update exist location	{"location_id":"","MAIN_CANTEEN","location_name":"","Main Cafeteria","capacity":"","1500"}	204 No Content	Pass	Pass	Pass	-	-
		TC_DBC_LOC_07	DELETE	/five-campus-hub/api/v1/locations/OPEN_CANTEEN	Delete exist location	N/A	204 No Content	Pass	Pass	Pass	-	-
	events	TC_DBC_EVT_01	GET	/five-campus-hub/api/v1/events	Get all events	N/A	200 OK	Fail	Pass	Pass	BUG_DBC_EVT_01	Cycle2: Issue fixed
		TC_DBC_EVT_02	GET	/five-campus-hub/api/v1/events/search?title=Vramaya	Get one event	N/A	200 OK	Blocked	Pass	Pass	-	New in Cycle 2 (Not available in Cycle 1)
		TC_DBC_EVT_03	POST	/five-campus-hub/api/v1/events	Create new event	{"event_title":"","Hackathon","event_date":"","2026-06-20","event_time":"","10:00:00","location_name":"","Main Cafeteria","club_name":"","Leo Club"}	201 Created	Fail	Fail	Pass	BUG_DBC_EVT_02	Cycle2: Missing 'event_date' fixed. New bug found Cycle3: Issue fixed
		TC_DBC_EVT_04	PUT	/five-campus-hub/api/v1/events/name/Viramaya	Update exist event	{"event_title":"","Medusa","event_date":"","2026-07-10","event_time":"","16:30:00","location_name":"","Caf Park","club_name":"","Students Media Club"}	204 No Content	Fail	Fail	Pass	BUG_DBC_EVT_03	Cycle2: Missing 'event_date' fixed. New bug found Cycle3: Issue fixed
		TC_DBC_EVT_05	DELETE	/five-campus-hub/api/v1/events/event_50d77b7d-4a49-8702-65acc1feb0f1	Delete exist event	N/A	204 No Content	Fail	Pass	Pass	BUG_DBC_EVT_04	Cycle2: Issue fixed
		TC_DBC_CLB_01	GET	/five-campus-hub/api/v1/clubs	Get all clubs	N/A	200 OK	Pass	Pass	Pass	BUG_DBC_CLB_01	Cycle2: Issue fixed
	clubs	TC_DBC_CLB_02	POST	/five-campus-hub/api/v1/clubs	Create new club	{"club_name":"","Rotaract Club","status":true}	201 Created	Fail	Pass	Pass	BUG_DBC_CLB_02	Cycle2: Issue fixed
		TC_DBC_CLB_03	PUT	/five-campus-hub/api/v1/clubs	Update exist club	{"club_id":"","club_3ba29d9d-af45-4234-896e-ac0add8a794","club_name":"","Badminton Club","status":true}	204 No Content	Pass	Pass	Pass	BUG_DBC_CLB_03	Cycle2: Issue fixed
		TC_DBC_CLB_04	PATCH	/five-campus-hub/api/v1/clubs/club_5156a309-7b0a-4b62-ba7b-8f094569aeb3?status=false	Update only club status	N/A	204 No Content	Fail	Fail	Fail	BUG_DBC_CLB_04	Used @PathVariable but no path is defined
		TC_DBC_CLB_05	DELETE	/five-campus-hub/api/v1/clubs/club_2009f273-723e-42f2-8dca-a5b4540a607b	Delete exist club	N/A	204 No Content	Pass	Pass	Pass	-	-
		TC_DBC_STC_01	GET	/five-campus-hub/api/v1/students/club	Get all student club records	N/A	200 OK	Fail	Pass	Pass	BUG_DBC_STC_01	Cycle2: Missing 'joined_date','active_status' fixed. Cycle3: New bug found
	students/club	TC_DBC_STC_02	POST	/five-campus-hub/api/v1/students/club/join	Create new student club record	{"student_id":"","STU_014","club_id":"","club_3ba254ad6-922a-449e-312b-8853a42b5e3","active_status":true}	201 Created	Fail	Pass	Pass	BUG_DBC_STC_02	Cycle2: Authentication error fixed. Cycle3: New bug found
		TC_DBC_STC_03	PUT	/five-campus-hub/api/v1/students/club/C002	Update exist student club record	{"student_id":"","52409165","club_id":"","C002","joined_date":"","2024-08-25","active_status":true}	204 No Content	Fail	Pass	Descoped	BUG_DBC_STC_03	Cycle2: Authentication error fixed. Cycle3: Completely removed
		TC_DBC_STC_04	DELETE	/five-campus-hub/api/v1/students/club/leave	Delete exist student club record	{"student_id":"","52409165","club_id":"","C002","joined_date":"","2026-02-04","active_status":true}	204 No Content	Fail	Pass	Pass	BUG_DBC_STC_04	Cycle2: Authentication error fixed.
		TC_DBC_STC_05	PATCH	/five-campus-hub/api/v1/students/club/STU_014/club_3ba254ad6-922a-449e-932b-8883a42b5e3?deactivate	Update student's active status	N/A	204 No Content	Blocked	Blocked	Pass	-	New in Cycle 3 (Not available in Cycle 1, 2)
		TC_DBC_BUS_01	GET	/five-campus-hub/api/v1/buses/get-all	Get all buses	N/A	200 OK	Pass	Pass	Pass	BUG_DBC_BUS_01	Cycle2: Missing fields added

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	Feature	Test ID	Test Scenarios	Expected Result	Actual Result	Cycle1 Status (Initial)	Cycle2 Status (Regression)	Cycle3 Status (Regression)	Bug ID	Notes	Improvement ID
				Pre Login Interface							
Branding	TC_UI_PRE_01	Campus name displays correctly	"CINEC"	"Kenton"	Fail	Pass	Pass	BUG_PRE_UI_01	-	-	
	TC_UI_PRE_02	Color palette matches branding	Navy blue + Sky blue	Navy blue + Gold	Fail	Pass	Pass	BUG_PRE_UI_02	-	-	
	TC_UI_PRE_03	Consistent component design	Consistent style across all components	Consistent	Pass	Pass	Pass	-	-	-	
Login Page	TC_UI_PRE_04	Login page loads correctly	All elements visible	All visible	Pass	Pass	Pass	-	-	UI_IMP_01	
	TC_UI_PRE_05	Seperate login form exist	Shows login form	Show login form	Pass	Pass	Pass	-	-	-	
	TC_UI_PRE_06	Student ID field validation	System blocks login and shows message like "Student ID required"	System blocks login and shows "Please enter your Student ID"	Pass	Pass	Pass	-	-	UI_IMP_02	
	TC_UI_PRE_07	Password field validation	System blocks login and shows messages like "Password required"	login success with empty, invalid password	Fail	Pass	Pass	BUG_PRE_UI_03	Cycle2: System blocks login and shows "Please enter both your Student ID and Password."	UI_IMP_03	
	TC_UI_PRE_08	Login with empty fields	System blocks login	System blocks login	Pass	Pass	Pass	-	-	-	
	TC_UI_PRE_09	Password Minimum character length	Shows "Password must be at least X characters"	No minimum character length	Fail	Fail	Deferred	BUG_PRE_UI_04	Postponed due to time limitation	-	
	TC_UI_PRE_10	Password Masking	Password characters hidden	Characters well hidden	Pass	Pass	Pass	-	-	-	
	TC_UI_PRE_11	Password visibility toggle	Password visible with eye icon	No visibility toggle	Fail	Fail	Deferred	BUG_PRE_UI_05	Postponed due to time limitation	-	
	TC_UI_PRE_12	Login successful feedback	Shows "Login Successful"	No feedback	Fail	Pass	Pass	BUG_PRE_UI_06	Cycle2: Shows "Login Successful"	UI_IMP_04	
	TC_UI_PRE_13	Back to home page button	Include back buttons to redirect home page	"Back to homepage" link available in login form	Pass	Fail	Pass	BUG_PRE_UI_07	Cycle2: "Back to homepage" link removed	UI_IMP_05	
	TC_UI_PRE_14	Link with registration page	Include a link to registration page	No link between login page and registration page	Fail	Pass	Pass	BUG_PRE_UI_08	Cycle2: Login page linked with registration page by "New here? Create new account"	-	
Sign Up Page	TC_UI_PRE_15	Seperate registration form exist	Shows sign up form	Open same login form with Student ID and Password fields	Fail	Pass	Pass	BUG_PRE_UI_09	Cycle2: Seperate registration form exist	UI_IMP_06	
	TC_UI_PRE_16	Full name field validation	System blocks registration and shows message like "Full name required"	No validation	Fail	Pass	Fail	BUG_PRE_UI_10	Cycle2: Name required to register	UI_IMP_07	
	TC_UI_PRE_17	Student ID field validation	System blocks login and shows message like "Student ID required"	No validation	Fail	Fail	Pass	BUG_PRE_UI_11	-	UI_IMP_08	
	TC_UI_PRE_18	Email field validation	System blocks login and shows message like "Email required"	No validation	Fail	Pass	Fail	BUG_PRE_UI_12	Cycle2: Email required to register	UI_IMP_09	
	TC_UI_PRE_19	Password field validation	System blocks login and shows message like "Password required"	No validation	Fail	Fail	Pass	BUG_PRE_UI_13	-	UI_IMP_10	
	TC_UI_PRE_20	Confirm Password field	Include "Confirm Password" field	No "Confirm Password" field	Fail	Fail	Deferred	BUG_PRE_UI_14	Postponed due to time limitation	-	
	TC_UI_PRE_21	Register with empty fields	System blocks registration	System blocks registration	Fail	Pass	Pass	BUG_PRE_UI_15	Cycle2: Registration blocks for empty fields	-	
	TC_UI_PRE_22	Password Minimum character length	Shows "Password must be at least X characters"	No minimum character length	Fail	Pass	Deferred	BUG_PRE_UI_16	Postponed due to time limitation	-	
	TC_UI_PRE_23	Password Masking	Password characters hidden	No password masking	Fail	Pass	Pass	BUG_PRE_UI_17	Cycle2: Password masked properly	-	
	TC_UI_PRE_24	Password visibility toggle	Password visible with eye icon	No visibility toggle	Fail	Fail	Deferred	BUG_PRE_UI_18	Postponed due to time limitation	-	
	TC_UI_PRE_25	Registration Successful feedback	Shows "Registration Successful"	No feedback	Fail	Pass	Pass	BUG_PRE_UI_19	Cycle2: Shows "Registration Successful"	UI_IMP_11	
	TC_UI_PRE_26	Back to home page button	Include back buttons to redirect home page	No back button	Fail	Fail	Pass	BUG_PRE_UI_20	-	UI_IMP_05	
	TC_UI_PRE_27	Link with login page	Include a link to login page	No link between registration page and login page	Fail	Pass	Pass	BUG_PRE_UI_21	Cycle2: Registration page linked with login page by "Back to login"	-	
Landing page	TC_UI_PRE_28	Cards positioning	Cards properly positioned, no overlap	Cards partially overlapping the background	Fail	Fail	Pass	BUG_PRE_UI_22	-	UI_IMP_12	
Responsive	TC_UI_PRE_29	Mobile view 375px	Layout adjusts properly	Have some wrapping issues between login and Sign up buttons	Fail	Pass	Pass	BUG_PRE_UI_23	Cycle2: Wrapping issue fixed	-	
	TC_UI_PRE_30	Tablet view 768px	Layout adjusts properly	Layout adjusts	Pass	Pass	Pass	-	-	-	
	TC_UI_PRE_31	Desktop view 1920 * 1080 px	Content centered properly	Content centered	Pass	Pass	Pass	-	-	-	
				Post Login Interface							
Dashboard	TC_UI_POST_01	Dashboard loads correctly	Dashboard visible with all core features	Loads and visible	Pass	Pass	Descoped	-	Cycle3: Completely removed	-	
	TC_UI_POST_02	User-friendly greeting	Greet with user name	Greet with student id	Fail	Pass	Descoped	BUG_POST_UI_01	Cycle2: Greets with name Cycle3: Completely removed	-	
	TC_UI_POST_03	Points balance displays	Shows points balance	Shows points balance	Pass	Pass	Descoped	-	Cycle3: Completely removed	-	
	TC_UI_POST_04	All the core features included	Main features including Transit, Events, Clubs, Checkin & Reward and Profile are clearly visible	Some features missing	Fail	Fail	Descoped	BUG_POST_UI_02	Cycle1: Missing Checkin, Rewards and Profile sections Cycle2: Added Checkin, Rewards and Profile but removed Events and Clubs pages Cycle3: Completely removed	-	
	TC_UI_POST_05	Quick actions	Quick action cards visible	No quick actions	Fail	Pass	Descoped	BUG_POST_UI_03	Cycle2: Added quick actions Cycle3: Completely removed	-	
	TC_UI_POST_06	Informative status bar	Status bar with colors showing live updates	Visible	Pass	Pass	Descoped	-	Cycle3: Completely removed		
	TC_UI_POST_07	Gamification elements	Include points balance, status indicators,	All visible	Pass	Pass	Descoped	-	Cycle3: Completely removed		
Navigation	TC_UI_POST_08	Navigation bar visibility	Shows links to main features	Shows links	Pass	Pass	Pass	-	Cycle3: Removed top nav bar and added side nav bar	UI_IMP_13	
	TC_UI_POST_09	Navigational consistency	Pages loads correctly	Pages loads	Pass	Pass	Pass	-	-	-	
	TC_UI_POST_10	Active menu highlight	Active menu section has different style	Active menu visible with a different style	Pass	Pass	Pass	-	-	UI_IMP_14	

[illegible]

Test Type	Feature	Bug ID	Bug Description	Test ID	Severity	Priority	Assign To	Status	Steps to Reproduce	Reported Date	Closed Date
Database & Backend Connection		BUG_DBC_DEP_01	In departments table, 'department_name' field update to unreadable text	TC_DBC_DEP_03	Critical	High	Backend Developer	Closed	1. Send POST request to /departments. 2. Verify HTTP response status code as 201 Created 3. Navigate to departments table in database 4. Verify department_id update to unreadable text	12/04/2026	25/05/2026
		BUG_DBC_STD_01	Student entity class and student table have 'enrolled_year' in different data types	TC_DBC_STD_01	High	High	Backend Developer	Closed	1. Send GET request to /students 2. Verify HTTP response as 500 Internal Server Error	31/03/2026	25/05/2026
		BUG_DBC_STD_02	Authentication error	TC_DBC_STD_02	High	High	Backend Developer	Closed	1. Send GET request to /students/me 2. Verify HTTP response as 500 Internal Server Error	12/04/2026	25/05/2026
		BUG_DBC_STD_03	Due to data type mismatch of 'enrolled_year' code tries to store NULL but DB rejects because of Not Null constraints	TC_DBC_STD_03	Critical	High	Backend Developer	Closed	1. Send POST request to /students 2. Verify HTTP response as 500 Internal Server Error	31/03/2026	25/05/2026
		BUG_DBC_STD_04	Authentication error	TC_DBC_STD_05	High	High	Backend Developer	Closed	1. Send PUT request to /students/me 2. Verify HTTP response as 500 Internal Server Error	12/04/2026	25/05/2026
		BUG_DBC_STD_05	Student table and student entity class have 'enrolled_year' in different data types. This cause for Delete requests	TC_DBC_STD_06	High	High	Backend Developer	Closed	1. Send DELETE request to /students/{id} 2. Verify HTTP response as 500 Internal Server error	31/03/2026	25/05/2026
		BUG_DBC_RWD_01	In rewards table, 'reward_id' field update to unreadable text	TC_DBC_RWD_03	Critical	High	Backend Developer	Closed	1. Send POST request to /rewards 2. Verify HTTP response status code as 201 Created 3. Navigate to rewards table in database 4. Verify reward_id update to unreadable text	12/04/2026	25/05/2026
		BUG_DBC_LOC_01	In locations table, 'location_id' field update to unreadable text	TC_DBC_LOC_05	Critical	High	Backend Developer	Closed	1. Send POST request to /locations 2. Verify HTTP response as 201 created 3. Navigate to locations table in database 4. Verify location_id update to unreadable text	12/04/2026	25/05/2026
		BUG_DBC_EVT_01	Failed to retrieve 'event_date' in event table due to missing 'event_date' attribute in Event entity class	TC_DBC_EVT_01	High	High	Backend Developer	Closed	1. Send GET request to /events 2. Verify status code as 500 Internal Server error	31/03/2026	25/05/2026
		BUG_DBC_EVT_02	Failed to create records in event table due to missing 'event_date' attribute in Event entity class	TC_DBC_EVT_03	High	High	Backend Developer	Closed	1. Send POST request to /events 2. Verify status code as 500 Internal Server error	31/03/2026	25/05/2026
	'club_id', 'event_title', 'location_id' store as null		Closed		1. Send POST request to /events 2. Verify status code as 201 Created 3. Check database 'club_id', 'event_title', 'location_id' columns	03/05/2026		25/05/2026			
		BUG_DBC_EVT_03	Failed to update records in event table due to missing 'event_date' attribute in Event entity class	TC_DBC_EVT_04	High	High	Backend Developer	Closed	1. Send PUT request to /events 2. Verify status code as 500 Internal Server error	31/03/2026	25/05/2026
	'club_id', 'event_title', 'location_id' update as null		Closed		1. Send POST request to /events 2. Verify status code as 204 No Content 3. Check database 'club_id', 'event_title', 'location_id' columns	03/05/2026		25/05/2026			
		BUG_DBC_EVT_04	Failed to delete records in event table due to missing 'event_date' attribute in Event entity class	TC_DBC_EVT_05	High	High	Backend Developer	Closed	1. Send DELETE request to /events 2. Verify status code as 500 Internal Server error	31/03/2026	25/05/2026
		BUG_DBC_CLB_01	Retrieved 'club_id' as null even though data available in database. Club table doesn't include 'student_id' as a FK, yet code tries to retrieve a 'student_id'	TC_DBC_CLB_01	High	High	Backend Developer	Closed	1. Send GET request to /clubs 2. Verify HTTP response status code as 200 OK 3. Verify club_id retrieve as NULL	08/04/2026	25/05/2026
		BUG_DBC_CLB_02	Couldn't create a new club record due to unnecessary student_id attribute	TC_DBC_CLB_02	Medium	Medium	Backend Developer	Closed	1. Send POST request to /clubs 2. Verify HTTP response status code as 500 Internal Server error	12/04/2026	25/05/2026
		BUG_DBC_CLB_03	In clubs table, 'club_name' update as NULL	TC_DBC_CLB_03	High	High	Backend Developer	Closed	1. Send PUT request to /clubs 2. Verify HTTP response status code as 204 No Content 3. Navigate to clubs table in database 4. Verify club_name update as NULL	12/04/2026	25/05/2026
		BUG_DBC_CLB_04	Couldn't update 'status' in clubs table because path variable for id in PATCH mapping not defined	TC_DBC_CLB_04	High	High	Backend Developer	Open	1. Send PATCH request to /clubs 2. Check curl controller 3. Observe error in console	12/04/2026	
		BUG_DBC_STC_01	"joined_date" and "active_status" attributes missing in StudentClub entity class	TC_DBC_STC_01	Medium	Medium	Backend Developer	Closed	1. Send GET request to /students/club 2. Verify HTTP response as 500 Internal Server Error	31/03/2026	25/05/2026
			"joined_date" returned as null		Open	1. Send GET request to /students/club 2. Observe joined date returned as null		21/05/2026			
		BUG_DBC_STC_02	Authentication error	TC_DBC_STC_02	High	High	Backend Developer	Closed	1. Send POST request to /students/club/join 2. Verify HTTP response as 500 Internal Server Error	12/04/2026	25/05/2026
			"joined_date" store as null		Open	1. Send POST request to /students/club/join 2. Verify HTTP response as 200 OK 3. Check database record		11/05/2026			
		BUG_DBC_STC_03	Authentication error	TC_DBC_STC_03	High	High	Backend Developer	Closed	1. Send PUT request to /students/club/COO2 2. Verify HTTP response as 500 Internal Server Error	12/04/2026	25/05/2026
		BUG_DBC_STC_04	Authentication error	TC_DBC_STC_04	High	High	Backend Developer	Closed	1. Send DELETE request to /students/club/leave 2. Verify HTTP response as 500 Internal Server Error	12/04/2026	25/05/2026
		BUG_DBC_BUS_01	In database, bus table missing 'bus_number' and 'status' fields	TC_DBC_BUS_01	Critical	High	Database Manager	Closed	1. Send GET request to /bus/getall 2. Verify HTTP response as 200 OK 3. Compare with Bus entity class	31/03/2026	25/05/2026
		BUG_DBC_BUS_02	In bus table, 'bus_id' field update to unreadable text	TC_DBC_BUS_02	Critical	High	Backend Developer	Closed	1. Send POST request to /bus/save 2. Verify HTTP status as 201 Created 3. Navigate to bus table in database 4. Verify bus_id update to unreadable text	12/04/2026	25/05/2026
			"status" field stored as null		Closed	1. Send POST request to /bus/save 2. Verify HTTP status as 201 Created 3. Check database status field stored as null		03/05/2026	25/05/2026		
		BUG_DBC_STR_01	Authentication error	TC_DBC_STR_01	High	High	Backend Developer	Closed	1. Send GET request to /student_rewards/admin/all 2. Verify HTTP response as 500 Internal Server Error	12/04/2026	25/05/2026
	BUG_DBC_STR_02	Authentication error	TC_DBC_STR_02	High	High	Backend Developer	Closed	1. Send GET request to /student_rewards/rewards 2. Verify HTTP response as 500 Internal Server Error	12/04/2026	25/05/2026	

UI Testing	Login functionality	BUG_LOG_STU_01	Student login with invalid details return 403 Forbidden instead of 401 Unauthorized	TC_LOG_STU_02	Low	Low	Backend Developer	Open	1. Send POST to /login with non-existent user id or invalid password 2. Observe 403 Forbidden instead of 401 unauthorized	02/05/2026		
				TC_LOG_STU_03								
		BUG_LOG_STU_02	Student login with empty details return 403 Forbidden instead of 400 Bad Request	TC_LOG_STU_04	Medium	Medium	Backend Developer	Open	1. Send POST to /login with empty user id and password 2. Observe 403 Forbidden instead of 400 Bad Request	02/05/2026		
				TC_LOG_STU_05								
				TC_LOG_STU_06								
		BUG_LOG_STU_03	Student login with SQL injection in fields return 403 Forbidden instead of 400 Bad Request	TC_LOG_STU_07	Low	Low	Backend Developer	Open	1. Send POST to /login with ' OR "1"="1" in id or password field 2. Observe 403 Forbidden instead of 400 Bad Request	02/05/2026		
				TC_LOG_STU_08								
				TC_LOG_STU_09								
		BUG_REG_STF_01	Staff login with invalid details return 403 Forbidden instead of 401 Unauthorized	TC_LOG_STF_02	Low	Low	Backend Developer	Open	1. Send POST to /login with non-existent user id or invalid password 2. Observe 403 Forbidden instead of 401 unauthorized	02/05/2026		
				TC_LOG_STF_03								
		BUG_REG_STF_02	Staff login with empty details return 403 Forbidden instead of 400 Bad Request	TC_LOG_STF_04	Medium	Medium	Backend Developer	Open	1. Send POST to /login with empty user id and password 2. Observe 403 Forbidden instead of 400 Bad Request	02/05/2026		
				TC_LOG_STF_05								
	TC_LOG_STF_06											
	BUG_REG_STF_03	Staff login with SQL injection in fields return 403 Forbidden instead of 400 Bad Request	TC_LOG_STF_07	Low	Low	Backend Developer	Open	1. Send POST to /login with ' OR "1"="1" in id or password field 2. Observe 403 Forbidden instead of 400 Bad Request	02/05/2026			
			TC_LOG_STF_08									
			TC_LOG_STF_09									
	Pre login Interface				TC_UI_PRE_01	High	High	Frontend Developer	Closed	1. Open main dashboard 2. Check campus name	03/04/2026	25/05/2026
					TC_UI_PRE_02	High	High	Frontend Developer	Closed	1. Open application 2. Check color palette used 3. Compare with Navy blue and Sky blue	03/04/2026	25/05/2026
					TC_UI_PRE_07	High	High	Frontend Developer	Closed	1. Open login page 2. navigate to login page 3. Fill only Student ID field 4. Click login	03/04/2026	25/05/2026
					TC_UI_PRE_09	Medium	Medium	Frontend Developer	Open	1. Open login page 2. Click on password field 3. Check if text shows minimum length	17/04/2026	
					TC_UI_PRE_11	Low	Medium	Frontend Developer	Open	1. Open login page 2. Check password field 3. Check for eye icon to showhide password	17/04/2026	
					TC_UI_PRE_12	Medium	Medium	Frontend Developer	Closed	1. Open login page 2. Enter valid details 3. Click login 4. Verify if message appears	17/04/2026	25/05/2026
					TC_UI_PRE_13	Low	Medium	Frontend Developer	Closed	1. Open login page 2. Check "Back to homepage" 3. Verify if its available	17/04/2026	25/05/2026
					TC_UI_PRE_14	High	High	Frontend Developer	Closed	1. Open login page 2. Check if "register here" link available	17/04/2026	25/05/2026
TC_UI_PRE_15					High	High	Frontend Developer	Closed	1. Open application 2. Check if Sign Up button opens separate registration page	17/04/2026	25/05/2026	
TC_UI_PRE_16					Medium	Medium	Frontend Developer	Open	1. Open registration form 2. Leave Full Name empty 3. Click register	17/04/2026		
TC_UI_PRE_17					High	High	Frontend Developer	Closed	1. Open registration form 2. Enter invalid student id 3. Click register	17/04/2026	25/05/2026	
TC_UI_PRE_18					High	High	Frontend Developer	Open	1. Open registration form 2. Enter invalid email 3. Click register	17/04/2026		
TC_UI_PRE_19					High	High	Frontend Developer	Closed	1. Open registration form 2. Leave password 3. Click register	17/04/2026	25/05/2026	
TC_UI_PRE_20					Medium	Medium	Frontend Developer	Open	1. Open registration form 2. Check if Confirm Password field exist	17/04/2026		
TC_UI_PRE_21					Critical	High	Frontend Developer	Closed	1. Open registration form 2. Leave all fields empty 3. Click register 4. Check if registration succeeds	17/04/2026	25/05/2026	
TC_UI_PRE_22					Medium	Medium	Frontend Developer	Open	1. Open registration form 2. Click on password field 3. Check if hint text shows minimum length	17/04/2026		
TC_UI_PRE_23					High	High	Frontend Developer	Closed	1. Open registration page 2. Type in password field 3. Check if characters show as dots or asterisks	17/04/2026	25/05/2026	
TC_UI_PRE_24					Low	Medium	Frontend Developer	Open	1. Open registration page 2. Look at password field 3. Check if eye icon	17/04/2026		
TC_UI_PRE_25					Medium	Medium	Frontend Developer	Closed	1. Open registration page 2. Enter valid details 3. Click register 4. Observe if success message appears	17/04/2026	25/05/2026	
TC_UI_PRE_26					Low	Medium	Frontend Developer	Closed	1. Open registration page 2. Look for "Back to home" or back button	17/04/2026	25/05/2026	
TC_UI_PRE_27					High	High	Frontend Developer	Closed	1. Open login page 2. Check if "Have a account? Login here" link available	17/04/2026	25/05/2026	
TC_UI_PRE_28					High	High	Frontend Developer	Closed	1. Open application 2. Look at hero section cards 3. Observe card positioning	03/04/2026	25/05/2026	
TC_UI_PRE_29					Medium	Medium	Frontend Developer	Closed	1. Open application on mobile view (375px) 2. Check login and sign up buttons 3. Identify wrapping issue	03/04/2026	25/05/2026	

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Source	Improvement ID	Sugestion	Current Behavior	Proposed Solution	Priority	Status	Reported Date	Reported By
UI Testing	UI_IMP_01	Fill background space in login page	Login page only have login form.	Use background image, gradient color or animation	Medium	Implemented	17/04/2026	QA Tester
	UI_IMP_02	Seperate validation message for Student ID field in login page	When login only with password, displays common message like "Please enter both your Student ID and Password."	Display seperate message like "Student ID required" or "Please enter your Student ID"	Medium	Proposed	17/04/2026	QA Tester
	UI_IMP_03	Seperate validation message for Password field in login page	When login only with Student ID, displays common message like "Please enter both your Student ID and Password."	Display seperate message like "Password required" or "Please enter your Password"	Medium	Proposed	17/04/2026	QA Tester
	UI_IMP_04	Display "Login Successful" feedback after login successfully	"Login Successful" message appear in top-right corner	Display the message in center to improve visibility	High	Proposed	17/04/2026	QA Tester
	UI_IMP_05	Include a back button to redirect to home page	Only browser back button available	Add a "Back to Home" icon to improve navigation flexibility	Medium	Implemented	17/04/2026	QA Tester
	UI_IMP_06	Name field improvement	Registration form only have full name field but database use seperate first name and last name columns	Remove Full Name field and add First Name and Last Name fields	High	Implemented	17/04/2026	QA Tester
	UI_IMP_07	Seperate validation message for Full Name field in registration page	When register with only other fields, displays common message like "Please enter a valid name and email"	Display seperate message like "Full Name required" or "Please enter your Full Name"	High	Proposed	17/04/2026	QA Tester
	UI_IMP_08	Seperate validation message for Student ID field in registration page	When register with only other fields, displays common message like "Please enter a valid name and email"	Display seperate message like "Student ID required" or "Please enter your Student ID"	High	Proposed	17/04/2026	QA Tester
	UI_IMP_09	Seperate validation message for Email field in registration page	When register with only other fields, displays common message like "Please enter a valid name and email"	Display seperate message like "Email required" or "Please enter your Email"	High	Proposed	17/04/2026	QA Tester
	UI_IMP_10	Seperate validation message for Password field in registration page	When register with only other fields, displays common message like "Please enter a valid name and email"	Display seperate message like "Password required" or "Please enter your Password"	High	Proposed	17/04/2026	QA Tester
	UI_IMP_11	Display "Registration Successful" feedback after register successfully	"Welcome to CINEC Hub, [Name]" message appear in top-right corner	Display the message in center to improve visibility	High	Proposed	17/04/2026	QA Tester
	UI_IMP_12	Landing page lacks strong visual engagement for first time users	Displays welcome message with brief description, navigation bar, login and sign up buttons	Enhance the hero section by adding CTA buttons, visua elements (illustration or image) and highlight main core features	High	Proposed	17/04/2026	QA Tester
	UI_IMP_13	Emojis in navbar look informal	Navbar use emojis to represent menu items	Replace emojis with a consistent icon set to improve usability and professionalism	Medium	Implemented	17/04/2026	QA Tester
	UI_IMP_14	Underline is not enough to notice active menu section	Active menu item use blue underline	Enhance by adding a background color or different text color for better visibility	Medium	Implemented	17/04/2026	QA Tester
	UI_IMP_15	Displaying multiple menu items reduce clarity	Same bottom navbar is used for both desktop and mobile view by containing multiple menu items	Use a top navbar for desktop view and keep bottom navbar only for core features on mobile view and move additional links using a hamburger menu	High	Implemented	17/04/2026	QA Tester
	UI_IMP_16	Display "Logout Successfully" feedback after logout successfully	"Logout Successfully" message appear in top-right corner	Display the message in center to improve visibility	High	Proposed	17/04/2026	QA Tester
	UI_IMP_17	Display Name, Email fields in profile section	Available only Name field	Add an Email field to retrieve and display user email	Medium	Implemented	17/04/2026	QA Tester
	UI_IMP_18	Use consistent font size for texts in buttons	Texts in buttons not clerly visible	Increase font size in all buttons for better visibility	High	Implemented	17/04/2026	QA Tester